Collaborative computing

Based on slides by Michel Beaudouin-Lafon (Univ. Paris Saclay)

Humans are social beings ...

Groups structure human activity

Professional life: teams, management chain,

Private life: family, friends, sport teams, choir, etc.

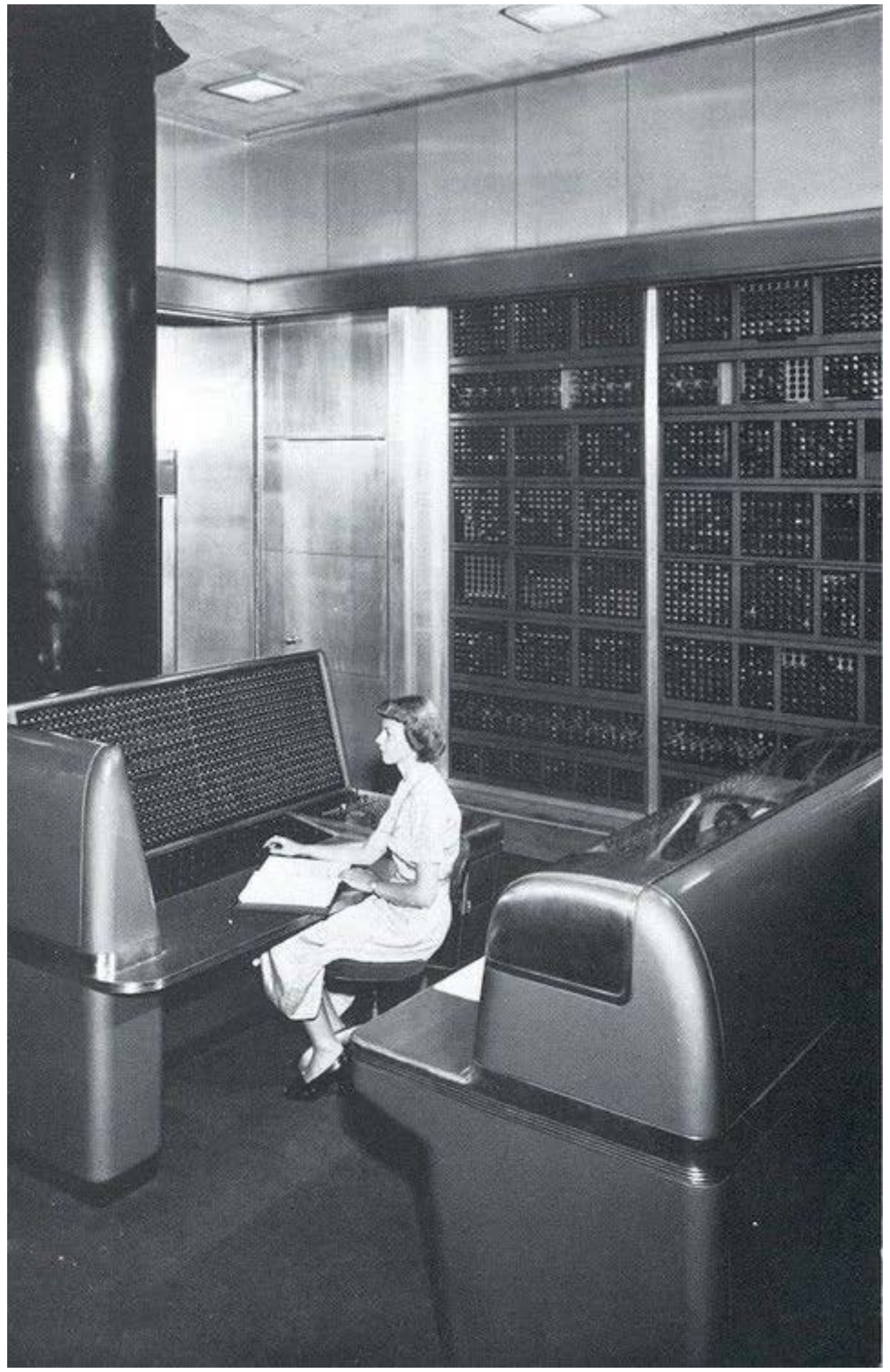
Groups are more than the sum of their parts

- Division of labor
- Take advantage of different expertise
- Transfer of skills: learning

... but computers are (mostly) personal

Time-sharing systems create the illusion that each user has access to all the resources and do not support awareness of what other users are doing

Example: file system



IBM SSEC, 1948

The PC is … the personal computer

Designed for
one user performing
one task with
one computer

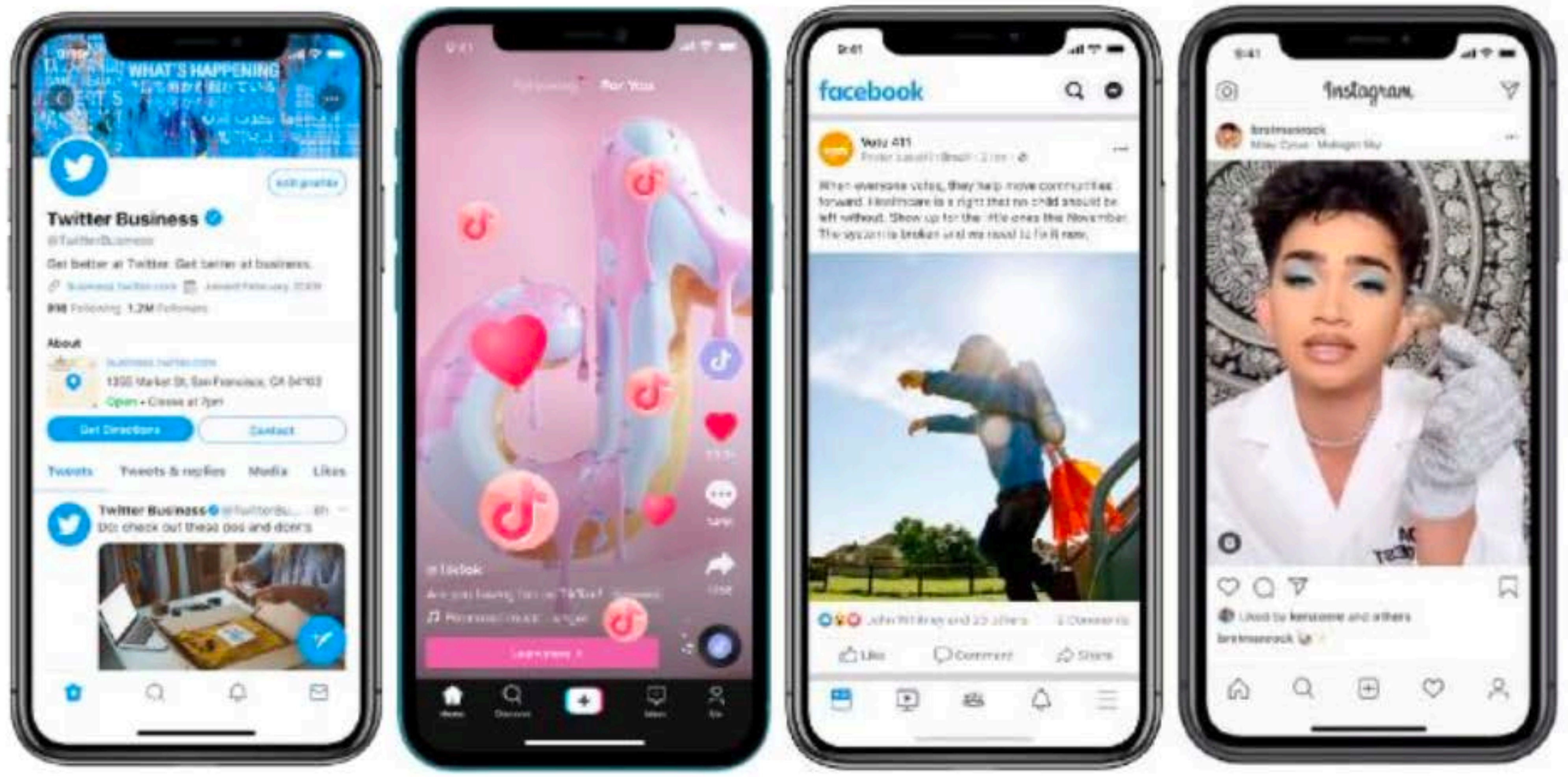
We live in a “connected” world

Or do we?

Trapped inside applications

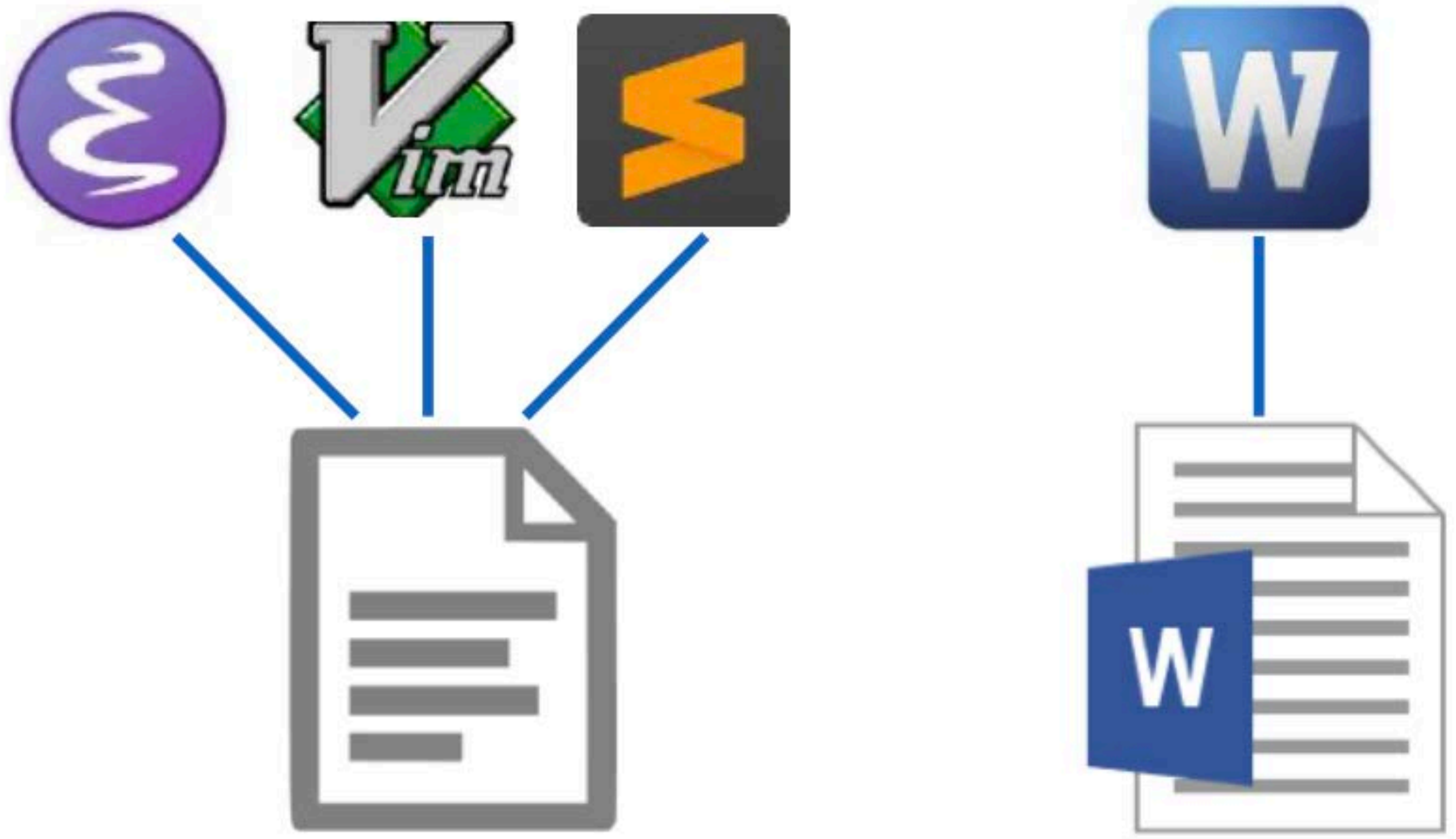
Email:
open and interoperable

Social networks: closed and
proprietary



Disowned of our files

My files:
personal control



The cloud:
sharing means giving away

No control over the means of communication

Telephone:
choice of operator and device

Videoconferencing: everybody must use the same service

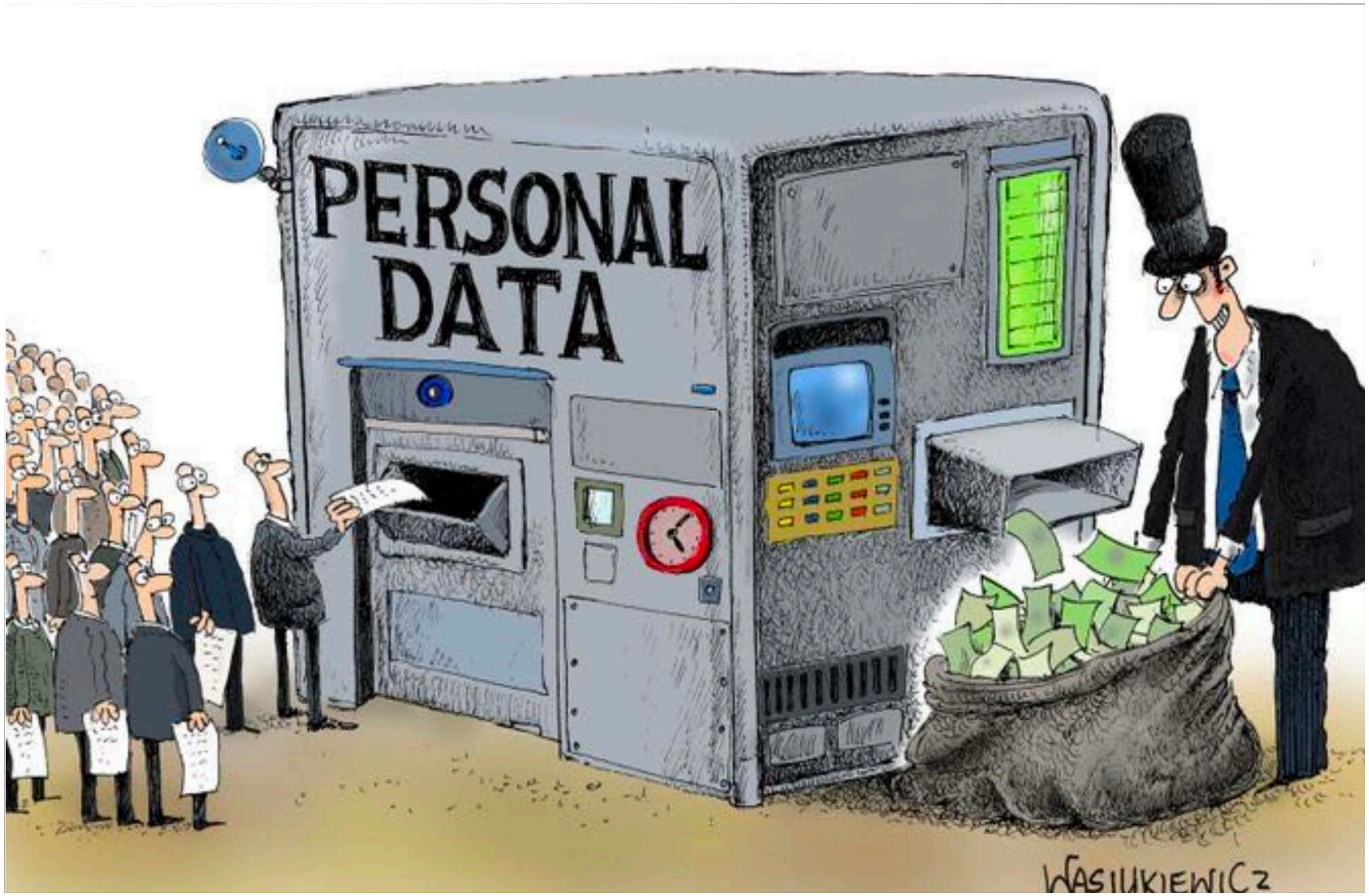
Walled gardens

We are trapped in
“app ecosystems”

Information silos

We don’t own our data

Why ?



Personal data collection

23	⌄	Ryanair	25.64%
24	⌄	Ocado	25.64%
25	^	Sleepcycle	25.64%
26	⌄	Paypal	25.64%
27	^	Google Maps	23.08%
28	NEW	YouTube	23.08%
29	NEW	Slack	23.08%
30	⌄	IKEA App	23.08%
31	⌄	Amazon	23.08%
32	⌄	Trainline	23.08%
33	⌄	Slimming World	23.08%
34	⌄	Walmart	23.08%
35	^	Whatsapp	20.51%
36	NEW	Zoom	20.51%
37	⌄	Deliveroo	20.51%
38	^	Protect Scotland	20.51%
39	^	CoStar	20.51%
40	⌄	CVS Pharmacy	20.51%
41	⌄	Offerup	20.51%
42	⌄	Doordash	20.51%
43	⌄	Amtrak	20.51%
44	NEW	Coinbase	17.95%
45	⌄	JustEat	17.95%
46	—	Facetune	15.38%
47	⌄	Wetherspoon	15.38%
48	⌄	Pornhub	15.38%
49	⌄	McDonalds (USA)	15.38%
50	⌄	Bet365 USA	15.38%
51	NEW	Headspace	12.82%
52	NEW	Google Docs	12.82%
53	NEW	Google Sheets	12.82%
54	NEW	Gmail	12.82%
55	NEW	VSCO	12.82%
56	⌄	Skybet	12.82%
57	⌄	Flo My Health	10.26%

Find the full report at **clario.co/blog/which-company-uses-most-data**

clario.

BUT:

Google reads all your email!

What can companies tell from image recognition?

The personal data that recognition software helps companies collect from you

Facial recognition
Recognises people and their key attributes

Background recognition
Detects elements in shot, establishes environment

Object recognition
Can identify an object or product within an image

#	Company	Face recognition	Environment recognition	Product recognition	Your contacts	Voice data/ recognition	Access to image library	Languages
1	Facebook	●	●	●	●	●	●	●
2	Instagram	●	●	●	●	●	●	●
3	TikTok	●	●	●	●	●	●	●
4	Twitter	●	●		●	●	●	●
5	Tinder	●	●		●		●	●

Find the full report at **clario.co/blog/which-company-uses-most-data**

clario.

Ad-based business model

Facebook income in 2020:

$84.2 billion - 98% from ads

$32.6 billion profit

38% profit margin

Facebook/Meta has about 3 billion users.

If each user paid $30 per year ($2.50 per month)
to be able to use Facebook, Instagram, WhatsApp,
Facebook/Meta would generate the same income

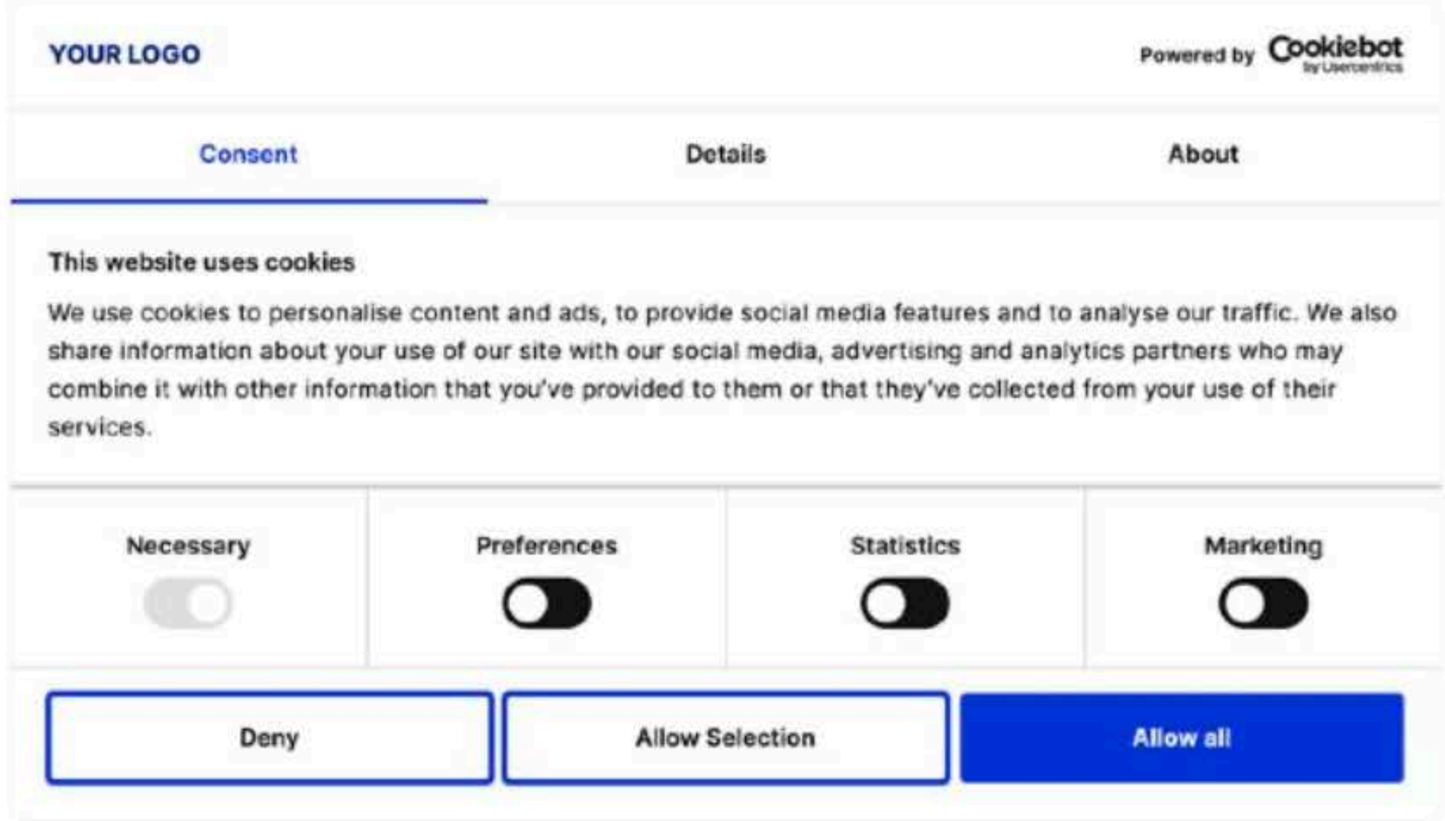
Apple (2020):

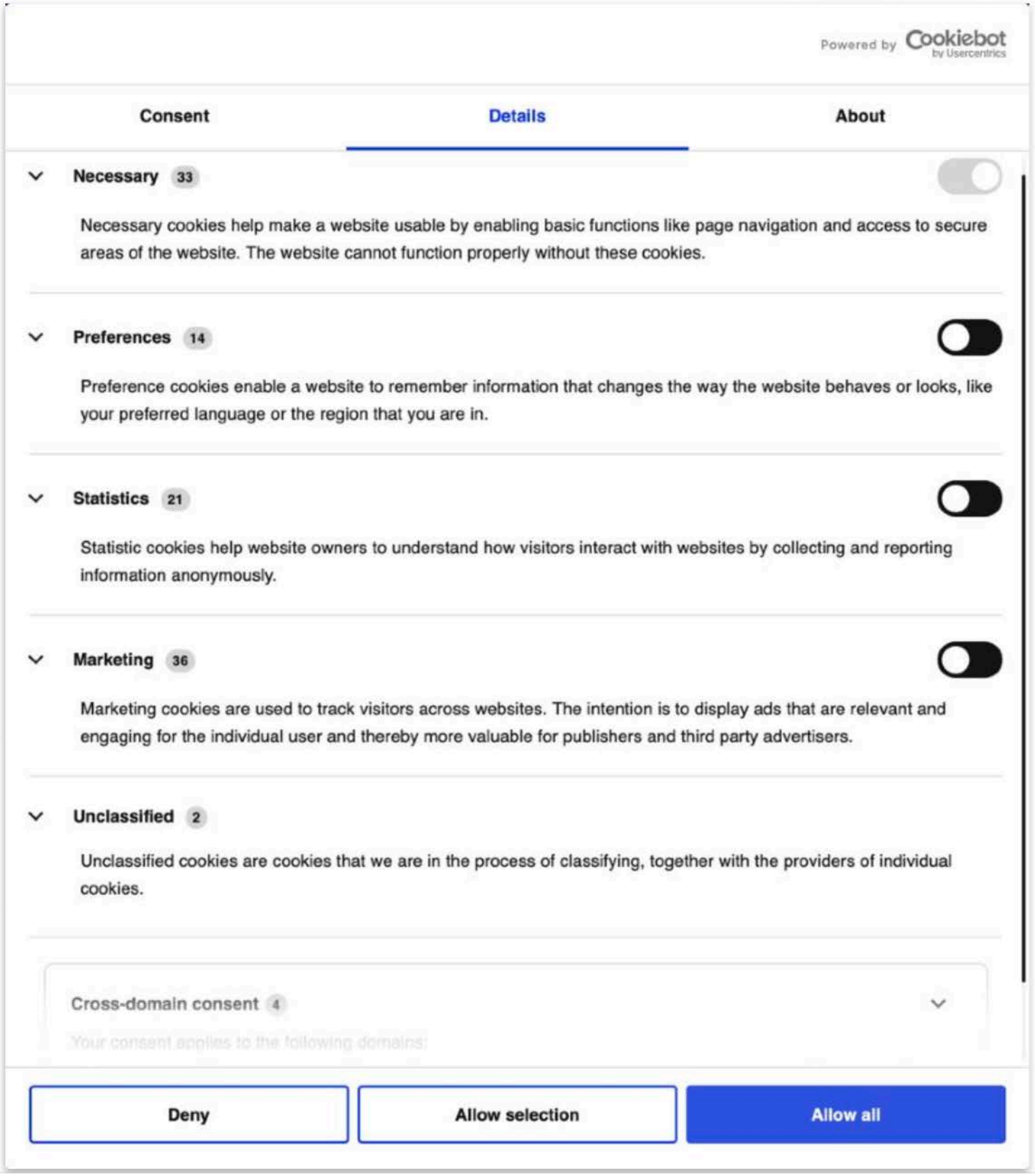
$209 billion income
$80 billion profit
38% profit margin

GDPR - General Data Protection Regulation

“Informed” consent

Do you know what you consent to?

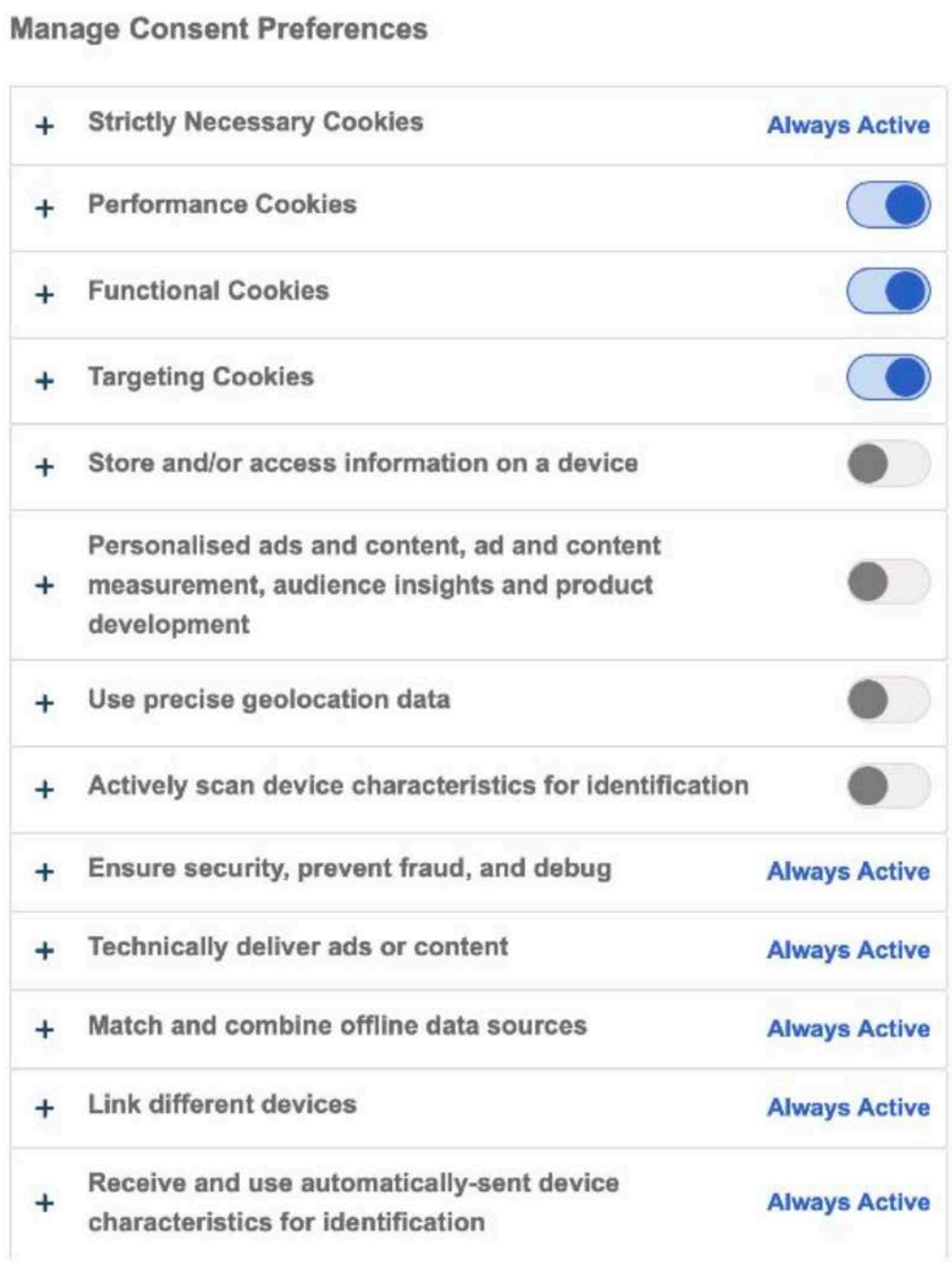




“Informed” consent

Example (investing.com):

more than 1600 vendors



Why is this relevant?

Collaborative computing
requires infrastructure to
mediate the collaboration

Whoever runs the infrastructure
controls
who can collaborate and how

This infrastructure becomes
a form of commons

Who should control it?

Under what rules?

What is Collaborative Computing?

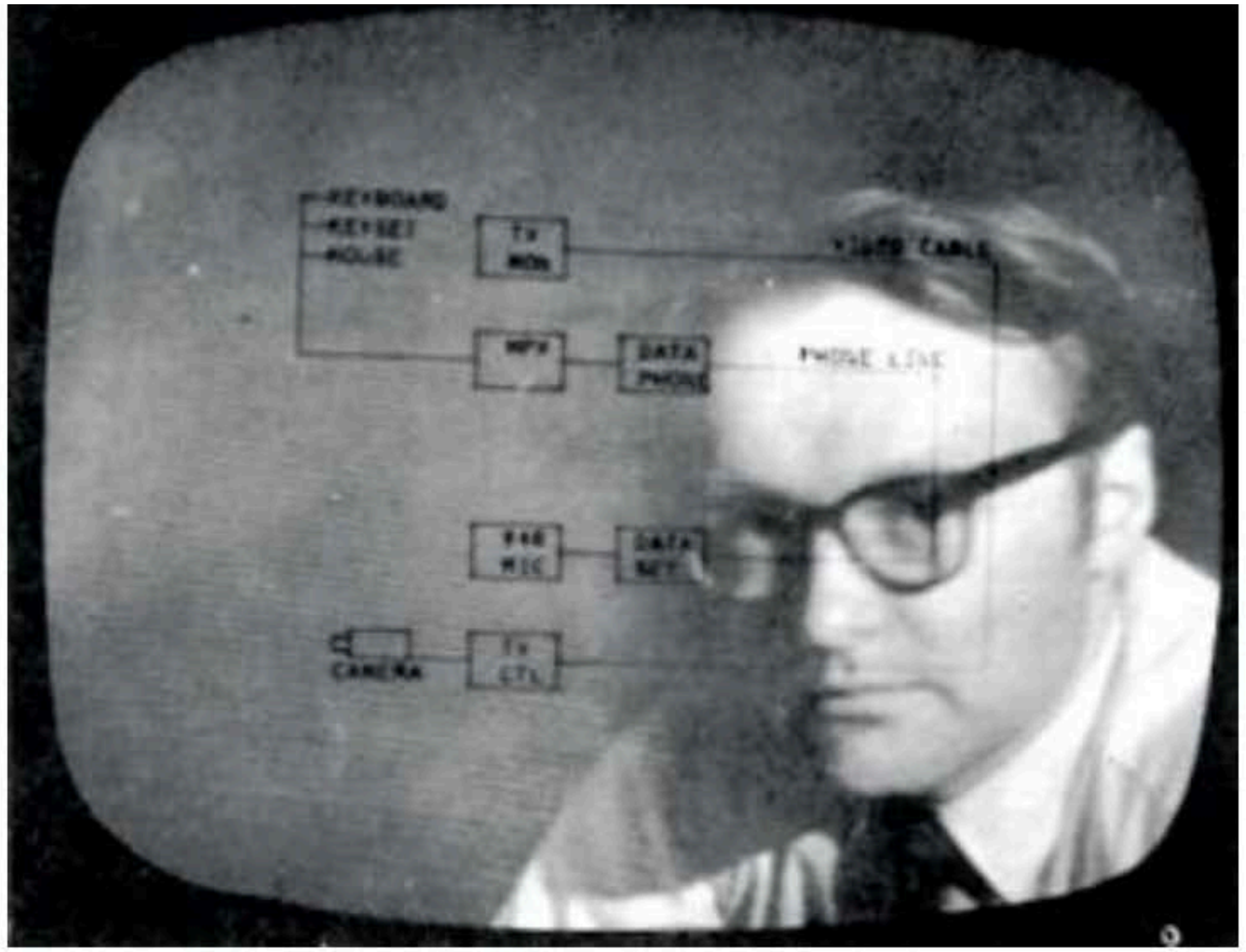
Don Norman

"Most work done on any complex entity is done by more than one person"

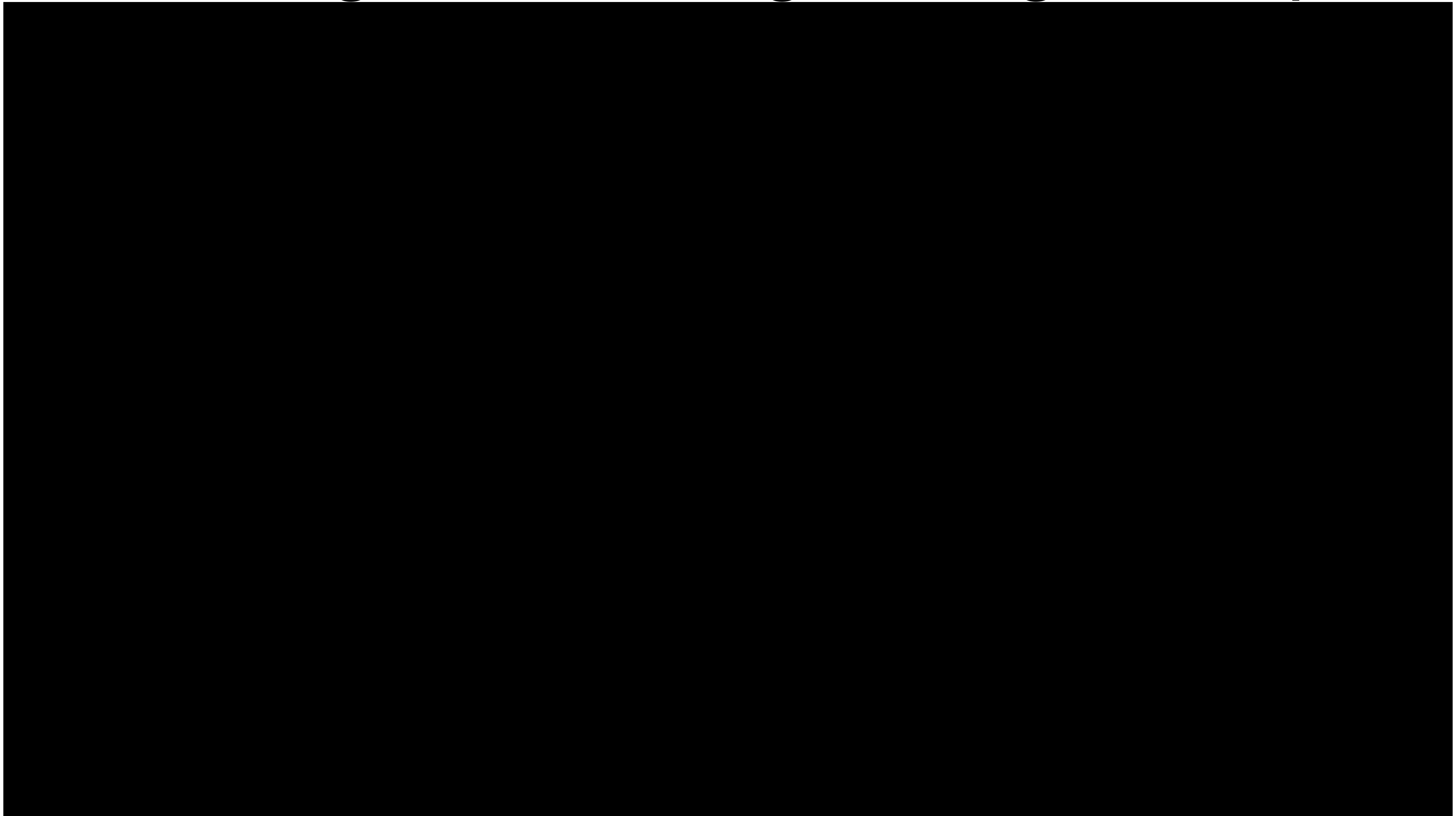
"Social impact of technology is hard to predict"

Augmenting the human intellect

1968 : Engelbart and his colleagues NLS/Augment, a system that supported file sharing, personal annotations, electronic messaging, videoconferencing, screen sharing, telepointers, etc.



NLS / Augment - Douglas Engelbart (1968)



Emergence of a field

Started under the umbrella of “Office Information Systems”
Software that supports group work
- Groupware (Johnson-Lenz, 1982)
- Computer Supported Cooperative Work (Greif & Cashman, 1984)

In French:
- *Collecticiel*
- *Travail Coopératif Assisté par Ordinateur (TCAO)*

Conferences: CSCW (ACM) and ECSCW since 1986
Journal of CSCW

Social definition

CSCW should be conceived as an endeavor to understand the nature and characteristics of cooperative work with the objective of designing adequate computer-based technologies. [...]

The focus is to understand, so as to better support, cooperative work.

Bannon et Schmidt, 1989

Engineering definition

Computer-based systems
that support
groups of people
engaged in
a common task (or goal)
and that provide
an interface to a shared environment

Ellis, Gibbs & Rein, 1991

Software definition

Groupware is distinguished from normal software
by the basic assumption it makes:

groupware makes the user aware that he is part of a group, while most other software seeks to hide and protect users from each other.

Lynch, Snyder & Vogel, 1990

Challenges

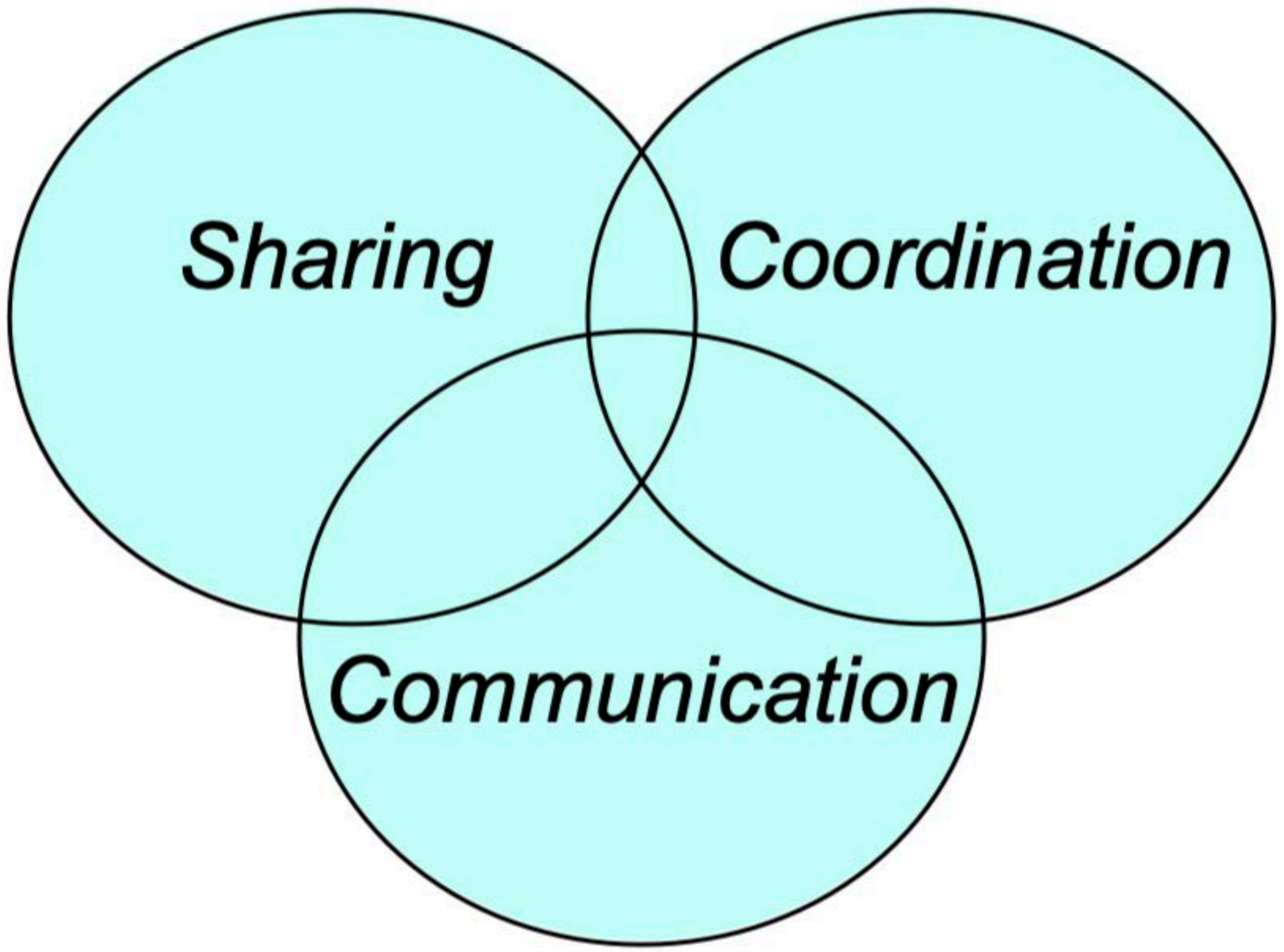
What should groupware systems do?
How to design them?
How do they affect use?

A multidisciplinary endeavor: sociology, ethnography, anthropology, design, computer science, etc.

Problems are both technical and human

Solutions are both technical and human

Functional taxonomy



Communication

exchanging information among participants

Sharing

accessing and editing digital artifacts

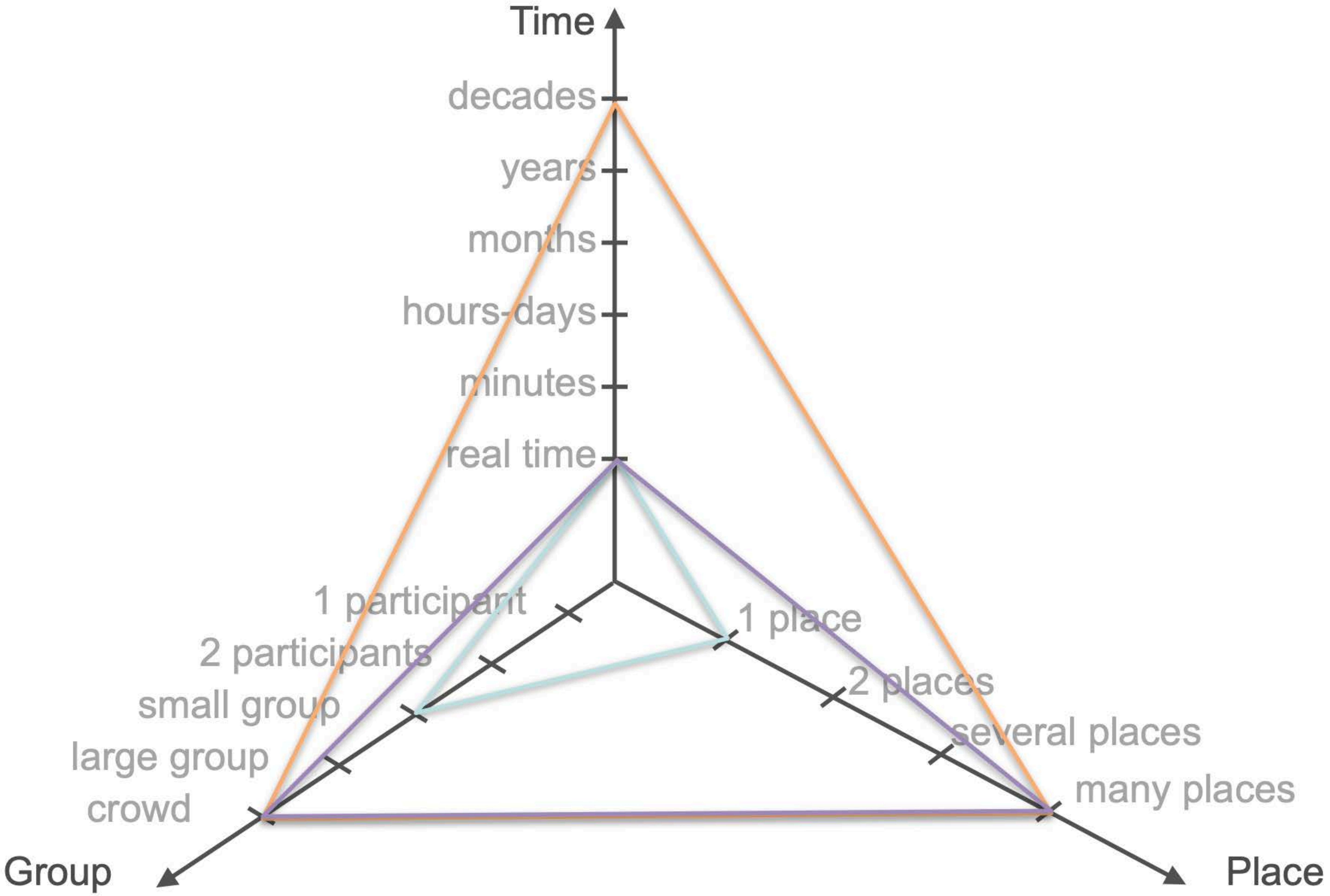
Coordination

division of labor among participants

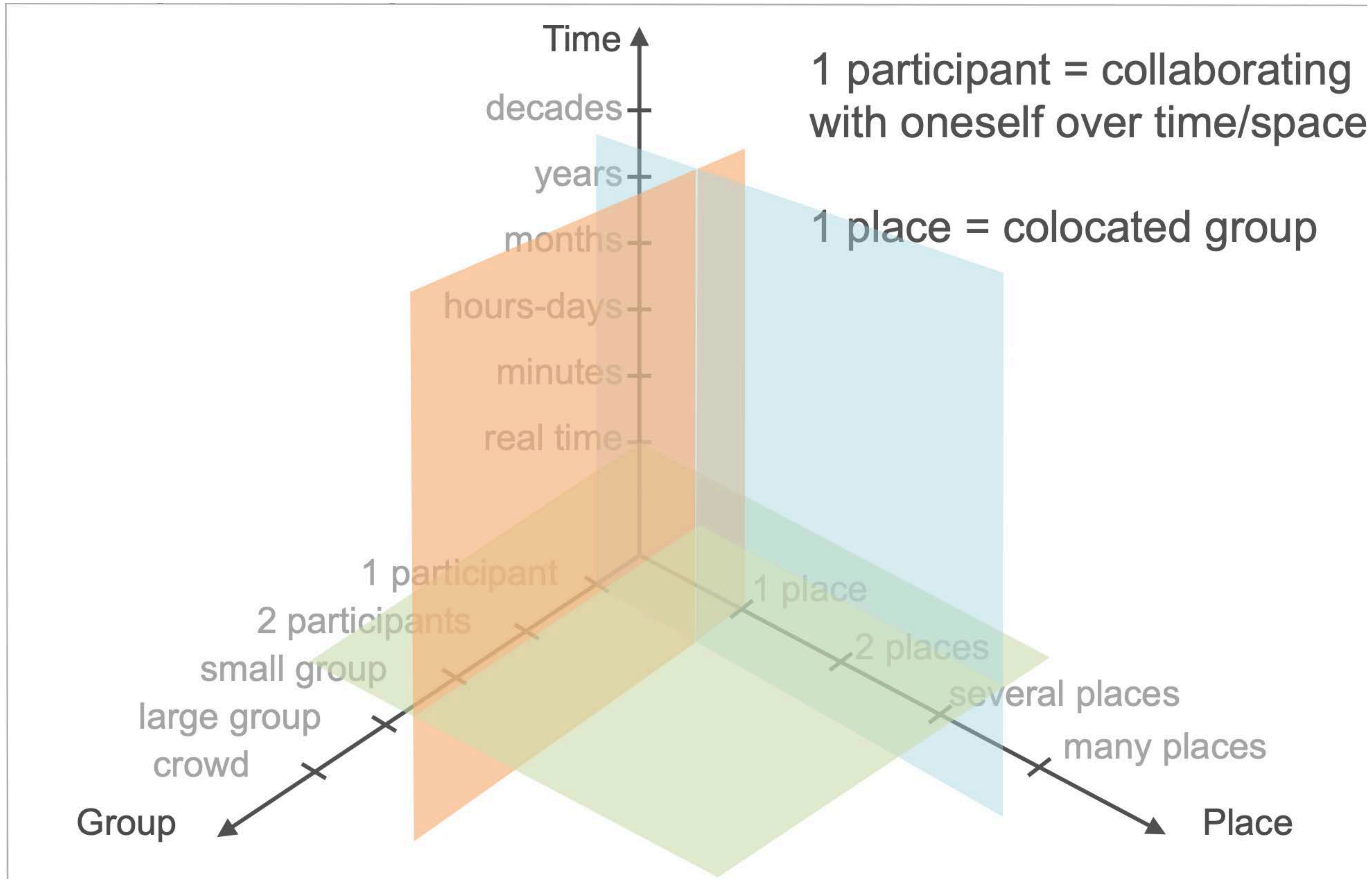
Space-time matrix

	Same place	Different place
Same time	face-to-face conversation	telephone call
Different time	Post-it note	letter

Group-time-space continuum



Group-time-space continuum



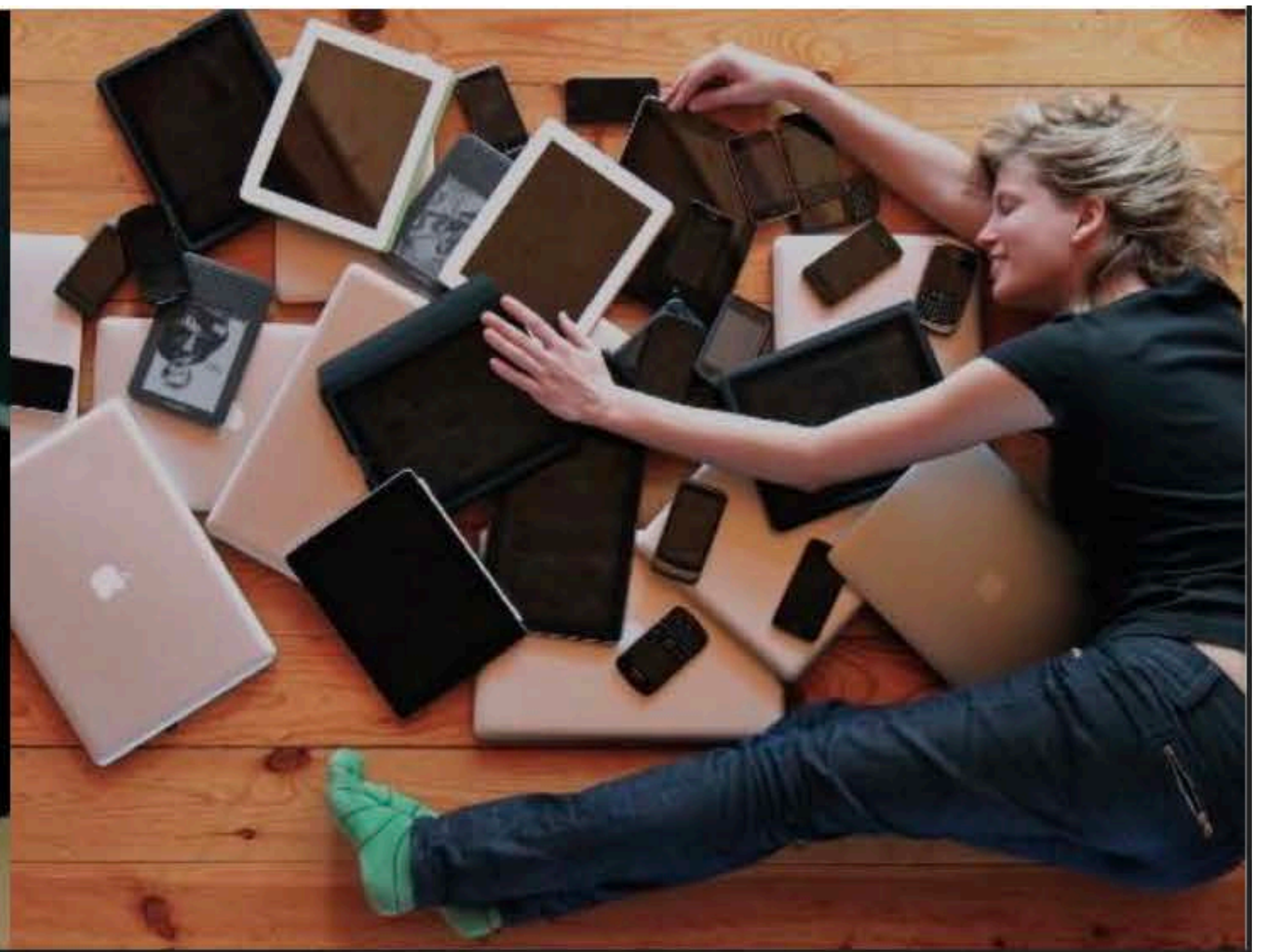
Scaling is difficult

Another dimension: devices

There are usually more devices than people

Collaboration with oneself across our devices

Making devices “collaborate” with each other



Collaboration is fluid

During the course of a collaborative activity,

the mode of collaboration changes, sometimes quickly

- from real time to different time
- from sharing to communicating to coordinating
- from tight coupling to loose coupling
- from everyone together to hybrid to fully remote

BUT most software do not address these transitions

A sample of collaborative systems

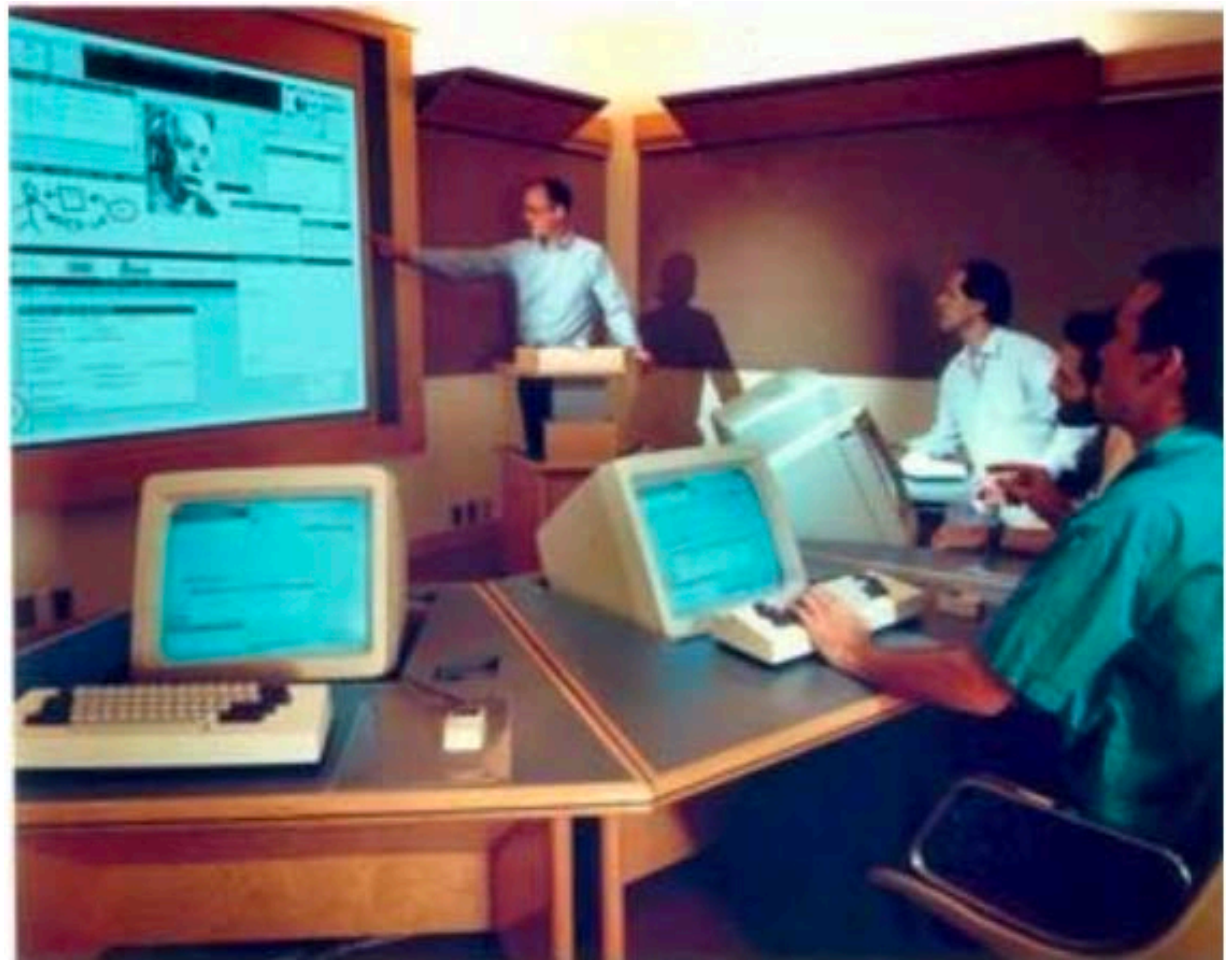
Share

Colab

Meetings of small group in a specially-equipped room

"Shared external memory"
- Boardnoter : hand drawing
- Cognoter : outlining ideas
- Argnoter : argumentation spreadsheet

View, space and time congruence
- What You See is What I See
- What You See Is Almost What I See



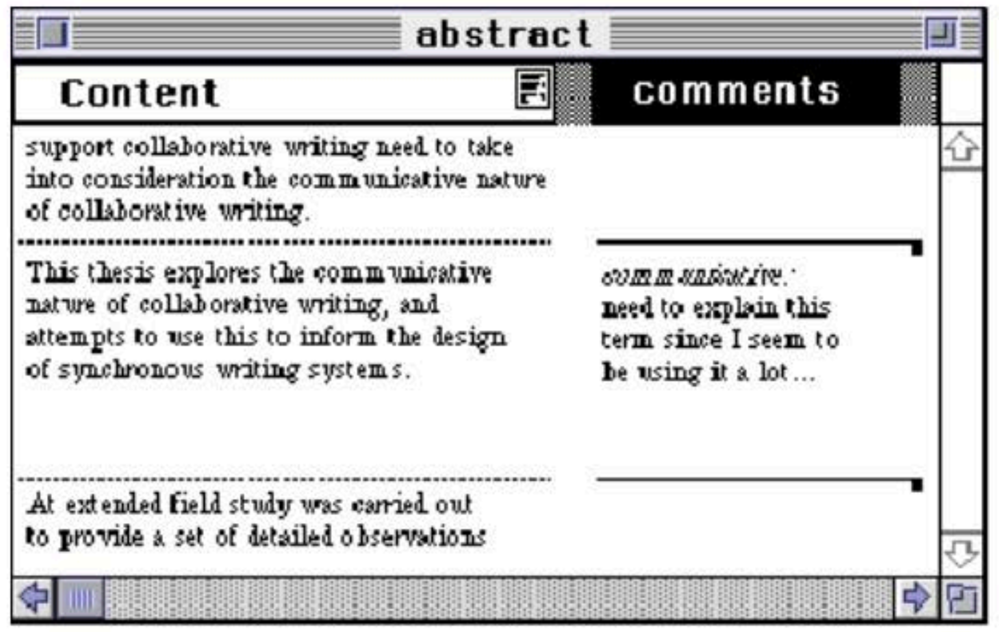
https://youtu.be/s2emx7qOqHU?si=VtoLL-Xnb2v1zYN5&t=278

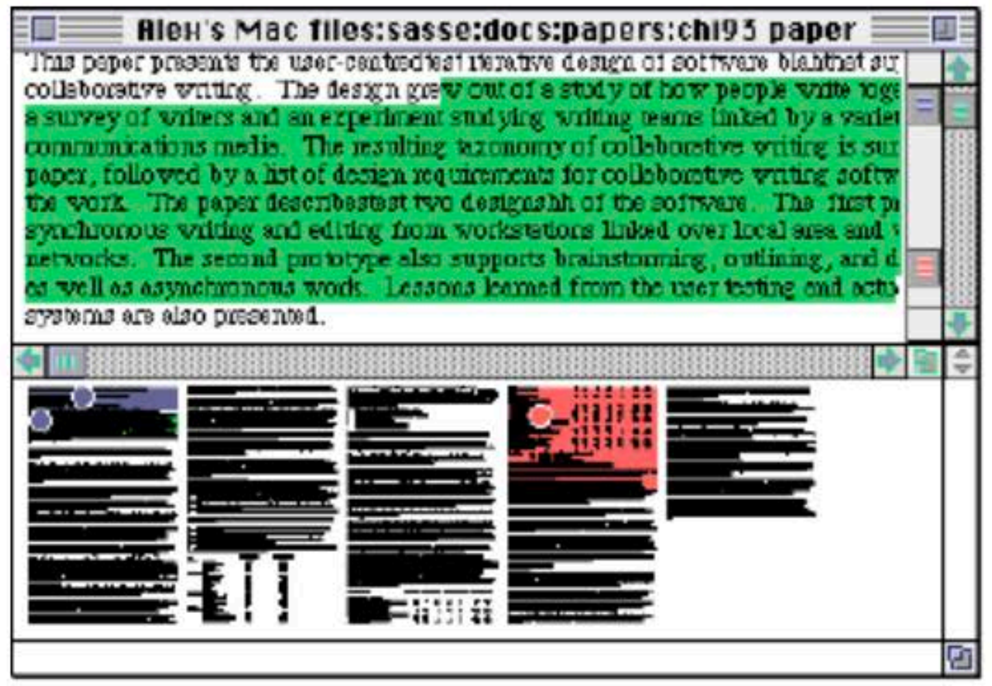
Shared editing

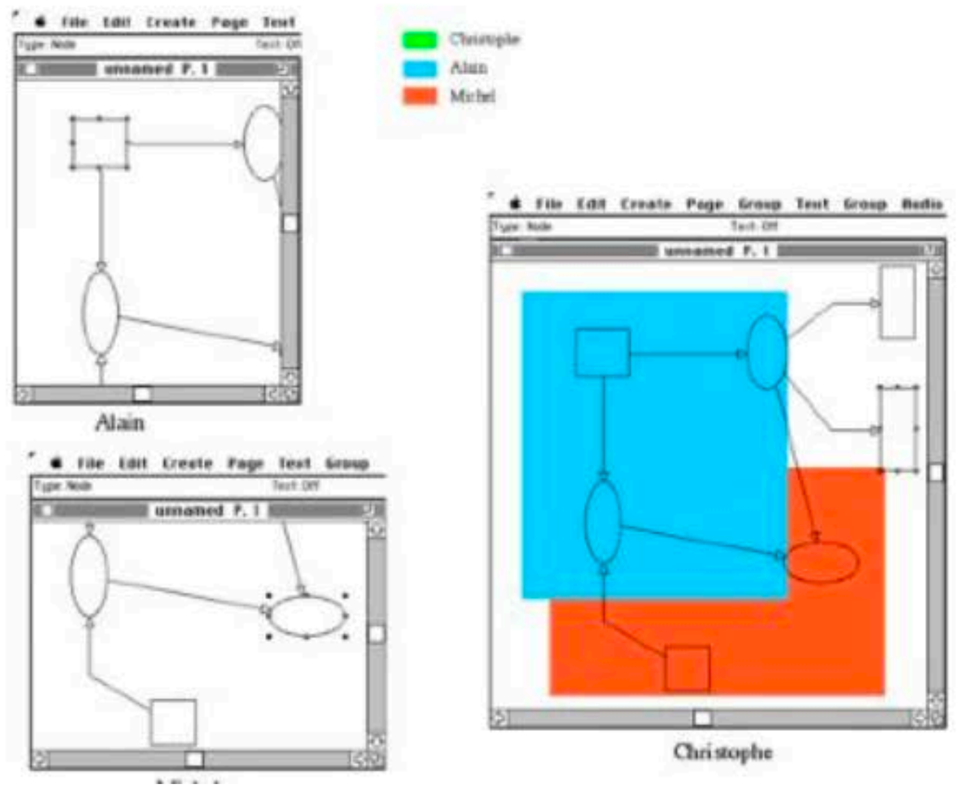
Text, asynchronous
– Quilt (Leland, Fish & Kraut, 1988)
– Prep (Neuwirth et al., 1989)

Text, synchronous
– Grove (Ellis, Gibbs & Rein, 1989)
– ShrEdit (McGuffin & Olson, 1992)
– SASSE (Baecker et al., 1993)

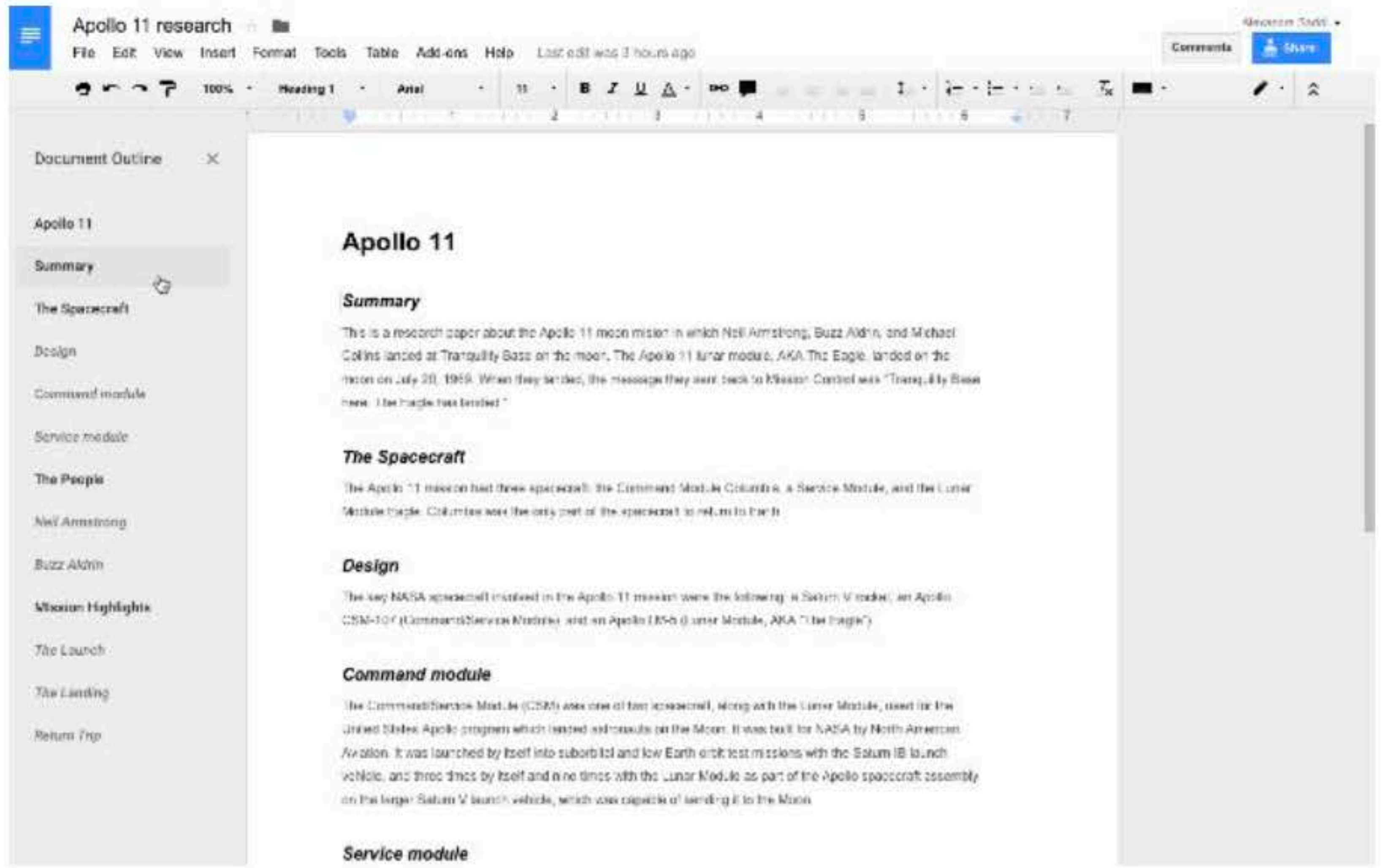
Graphics, synchronous
– GroupDesign
(Karsenty & Beaudouin-Lafon, 1992)



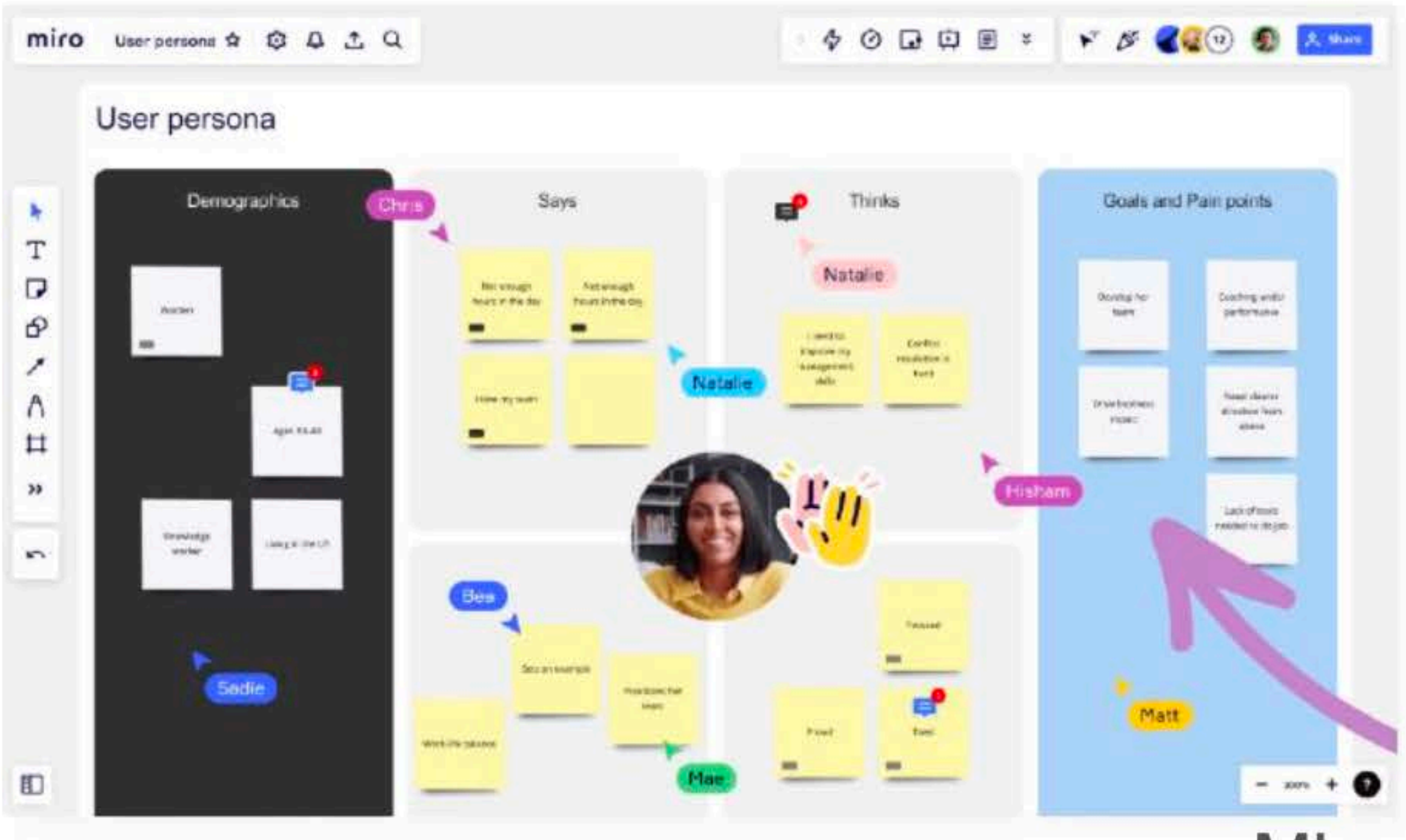




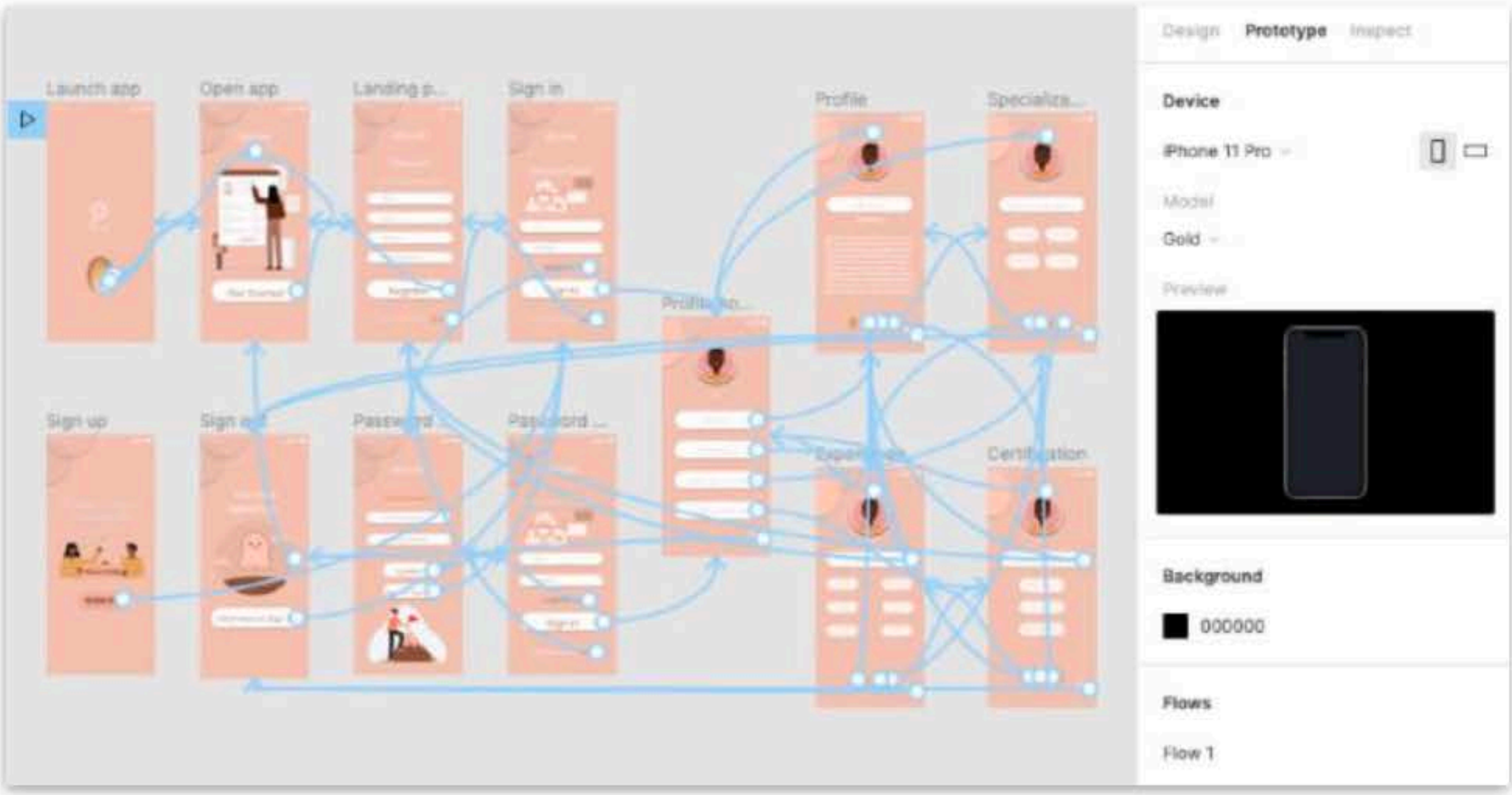
Web-based shared editing are now common



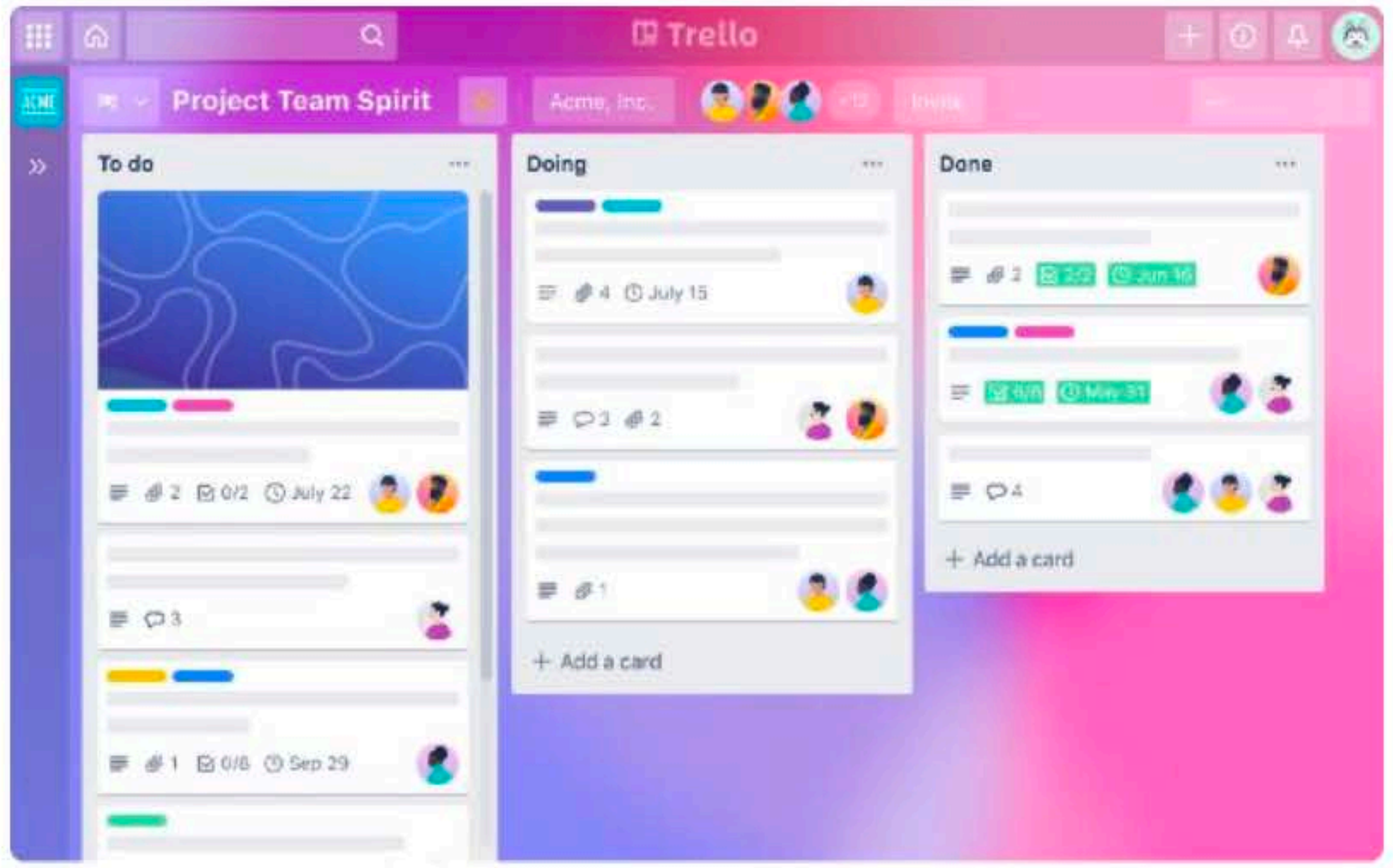
Google docs



Miro

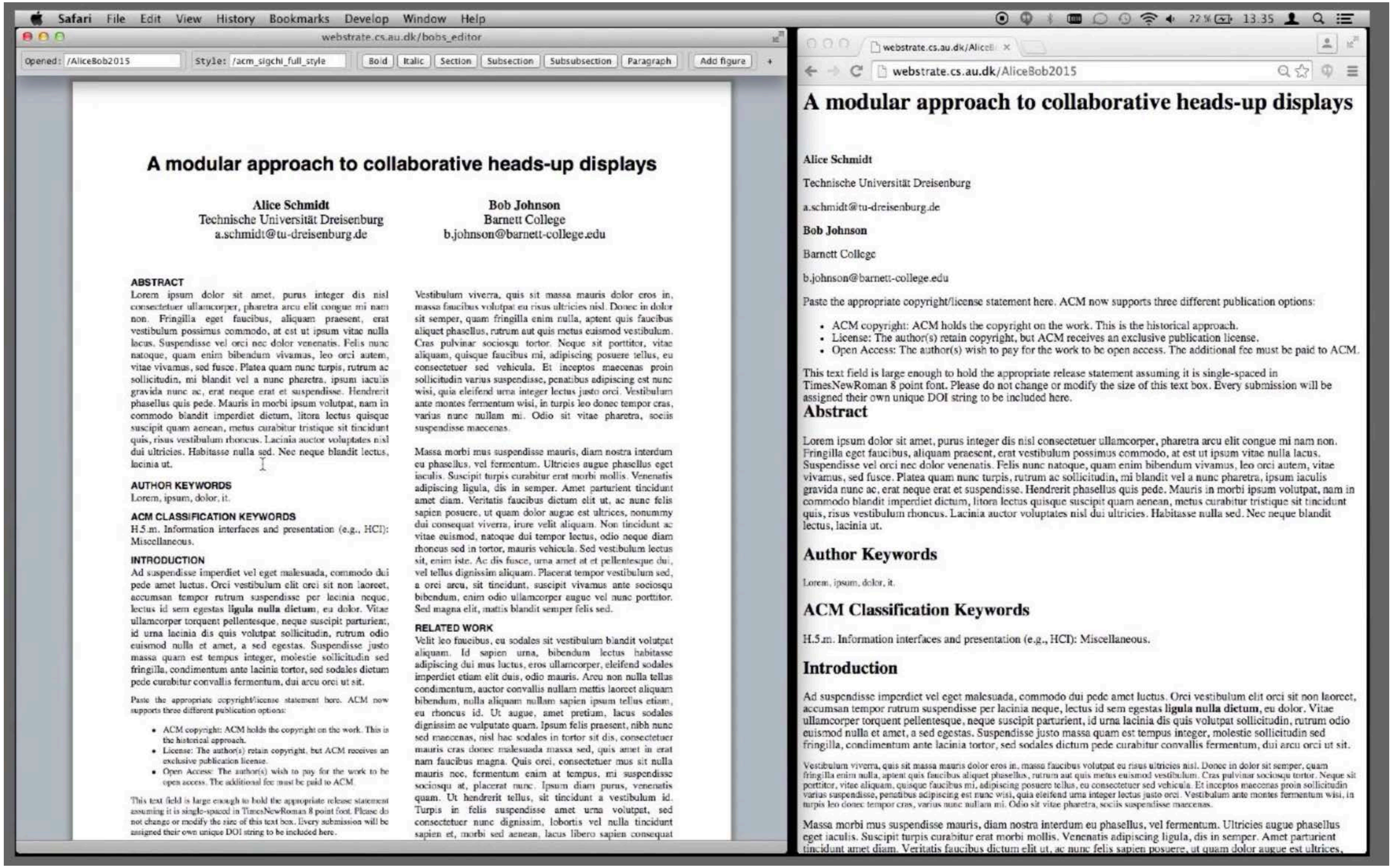


Figma



Trello

Webstrates


Safari File Edit View History Bookmarks Develop Window Help
webstrate.cs.au.dk/bobs_editor
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Subsection
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Paragraph
Add figure
A modular approach to collaborative heads-up displays
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ABSTRACT
AUTHOR KEYWORDS
Lorem, ipsum, dolor, it.
ACM CLASSIFICATION KEYWORDS
H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.
INTRODUCTION
RELATED WORK
webstrate.cs.au.dk/AliceBob2015
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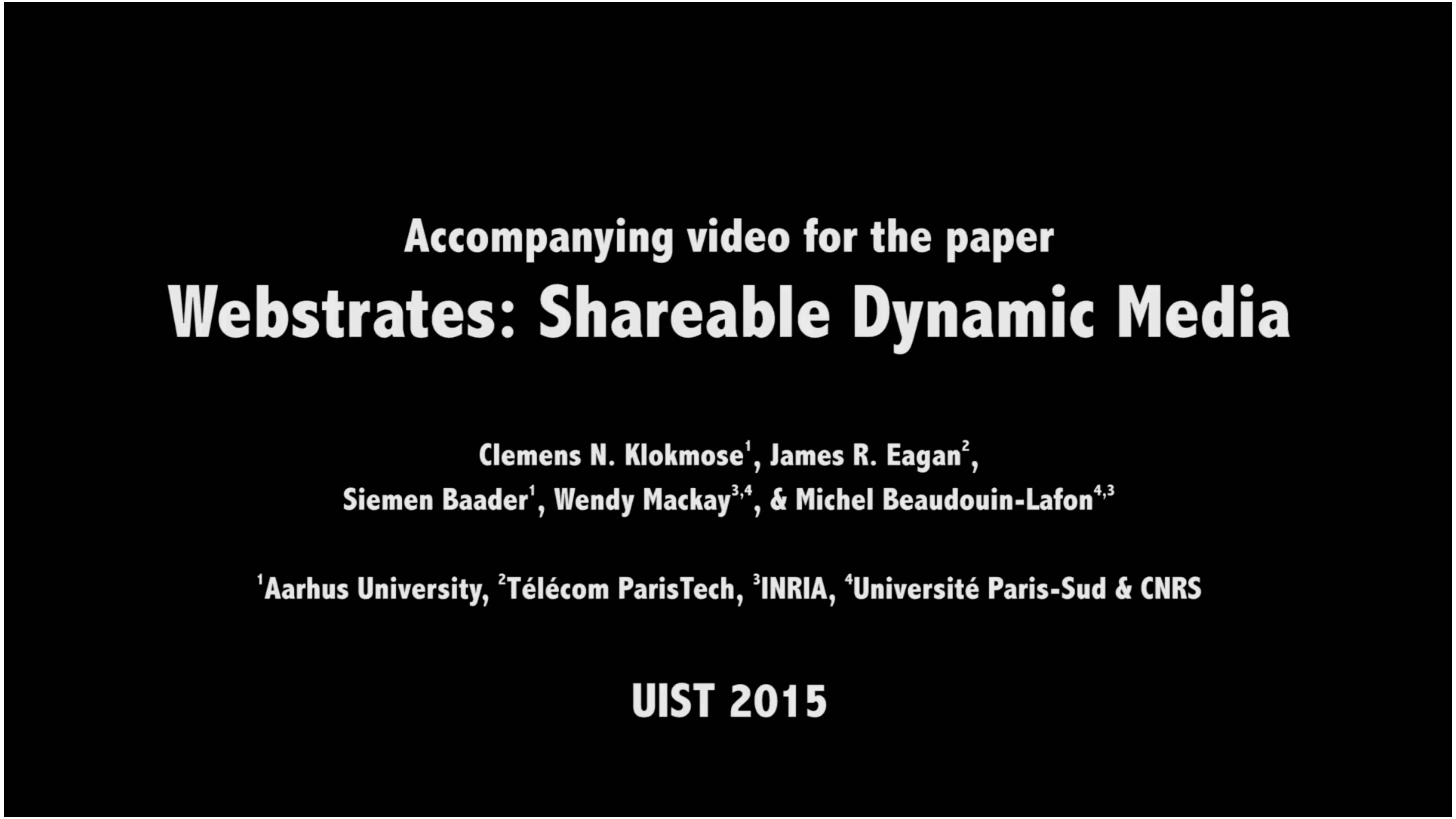
Abstract
Author Keywords
Lorem, ipsum, dolor, it.
ACM Classification Keywords
H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.
Introduction

Webstrates



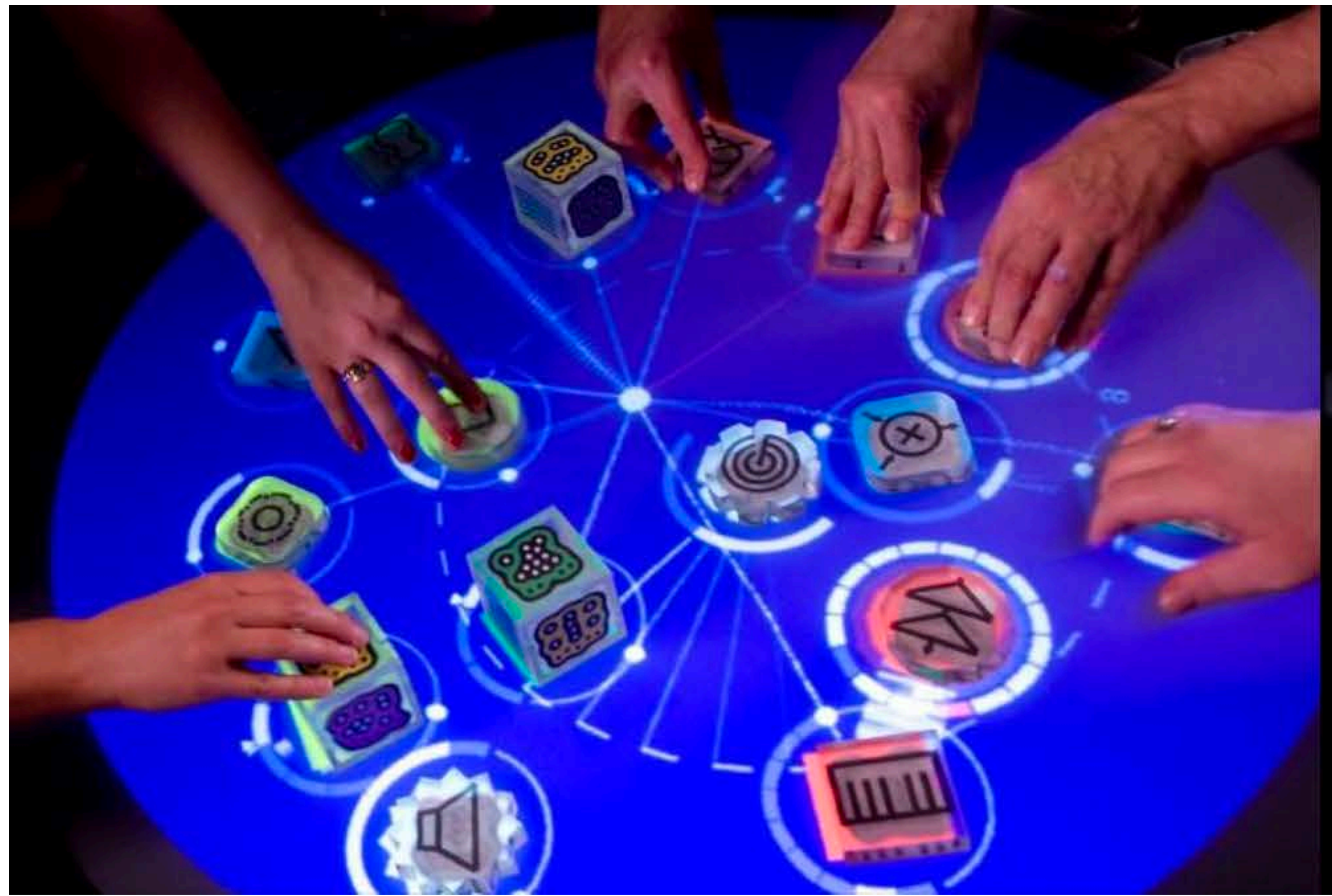
Accompanying video for the paper
Webstrates: Shareable Dynamic Media
Clemens N. Klokmose[1], James R. Eagan[2],
Siemen Baader[1], Wendy Mackay[3,4], & Michel Beaudouin-Lafon[4,3]
[1]Aarhus University, [2]Télécom ParisTech, [3]INRIA, [4]Université Paris-Sud & CNRS
UIST 2015

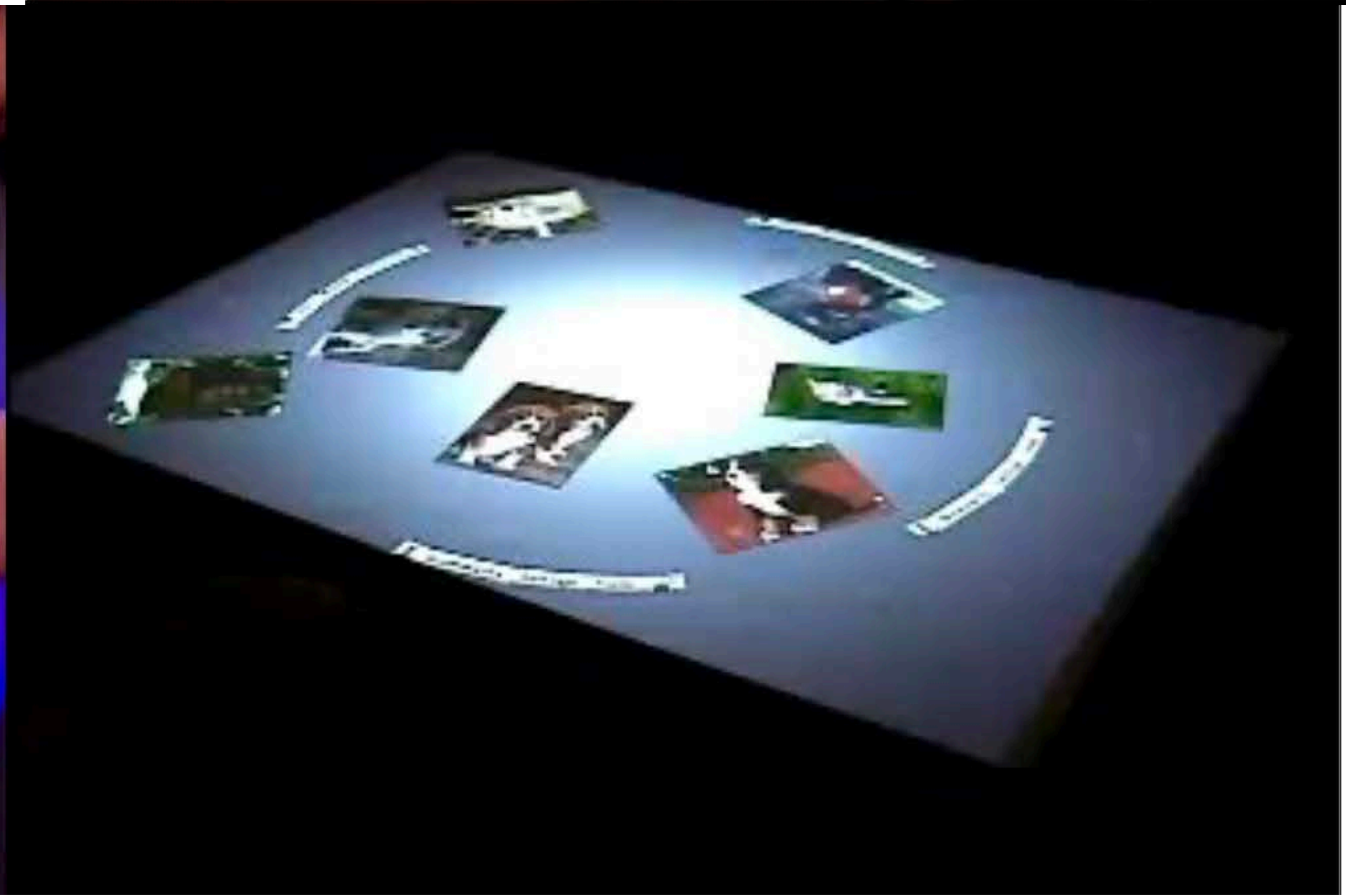
Tabletop interaction

DiamondTouch (Dietz et al., 2001)
- Interactive multitouch table
- Participant identification

DiamondSpin (Shen et al., 2004)
- User interface toolkit for tabletops

Reactable (Jordá et al., 2005)
- tangible interaction





Coordinate

Workflow systems

Managing a document across an organization
Example : a document includes metadata describing its path through an organization

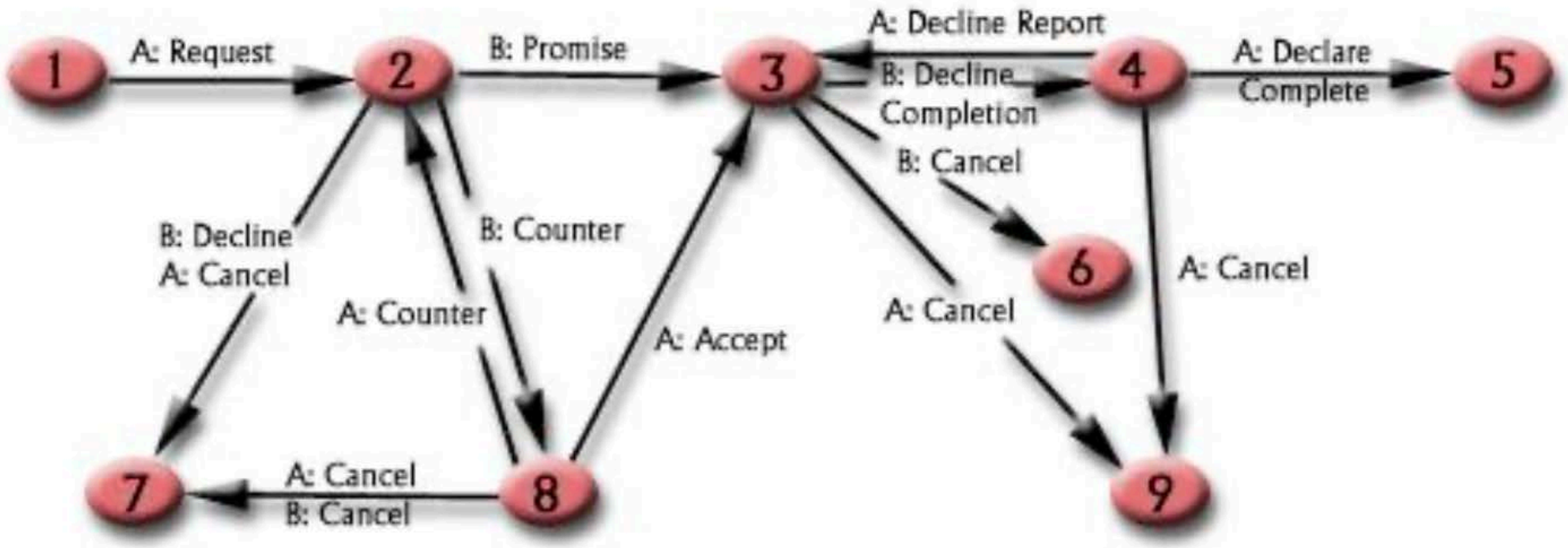
- must be written by Anne by April 15
- must be proofread by Bob by April 22
- must be approved by Charlie by April 29
- must be sent to Charlie by May 4

The document "knows its way"
and can send reminders
to the various people involved

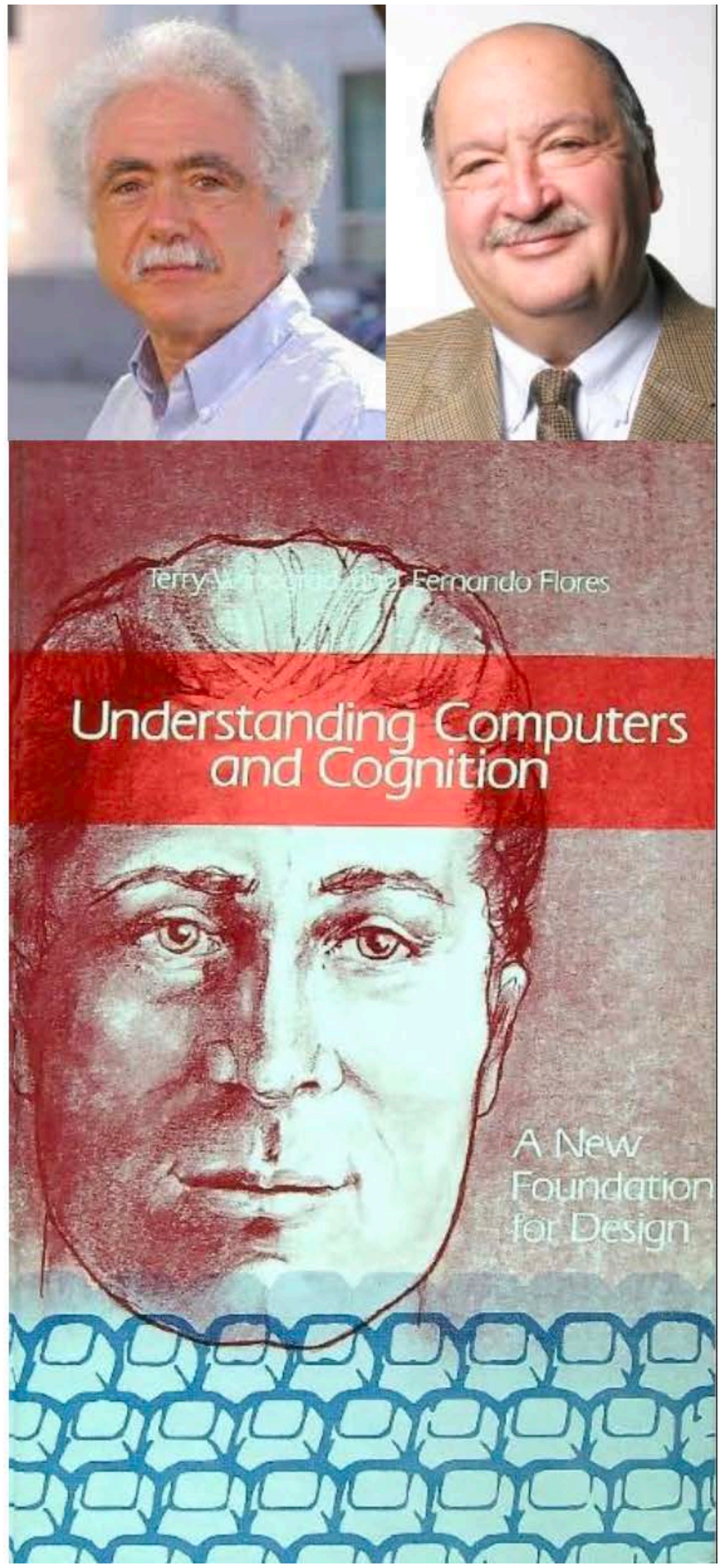
The Coordinator

Winograd & Flores, 1988

Based on the theory of speech acts (Searle)



Critique by Lucy Suchman

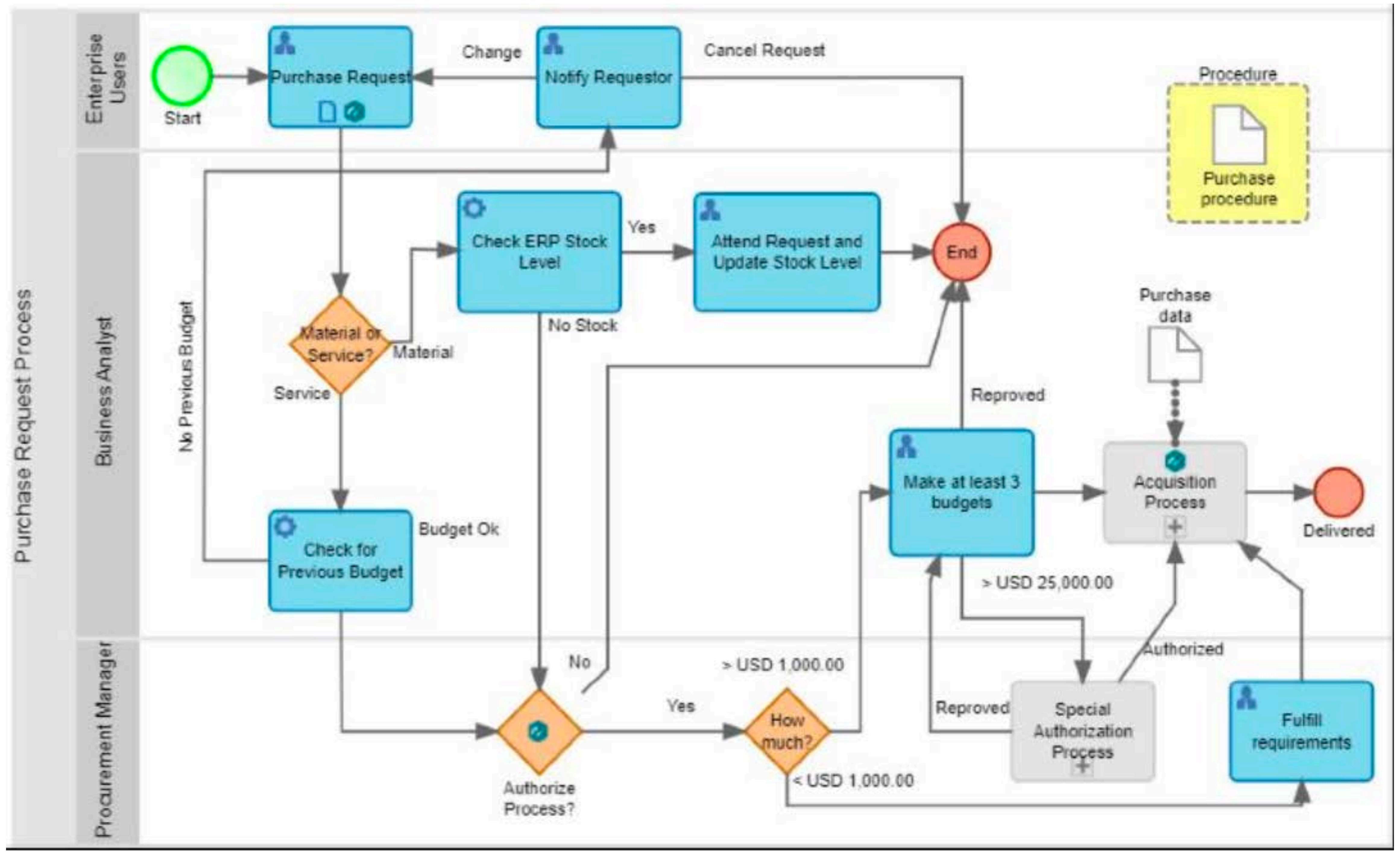


Business Process Management (BPM)

Optimize enterprise processes
Automate where possible

Large companies: SAP, SalesForce

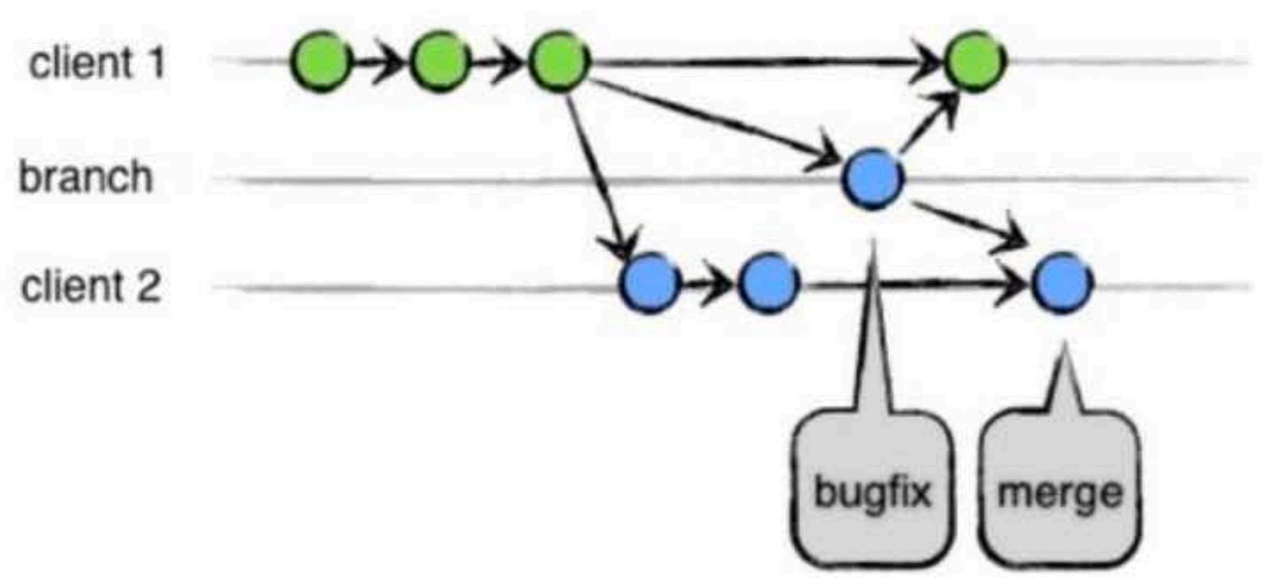
Modern form of Taylorism:
  Humans must conform
  to the processes



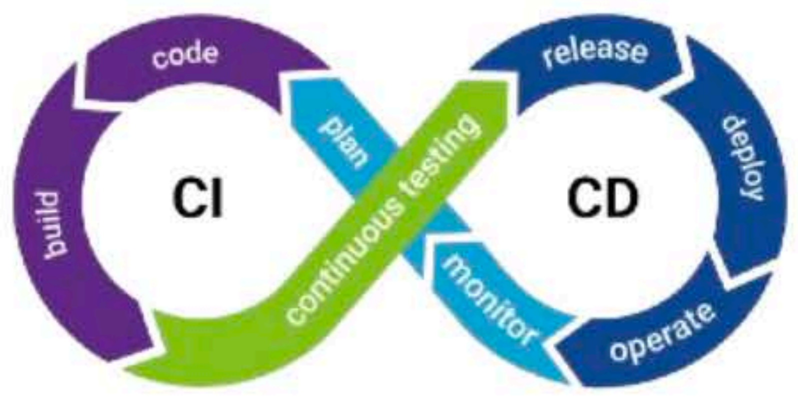
Collaborative software development

Manage complex software system development

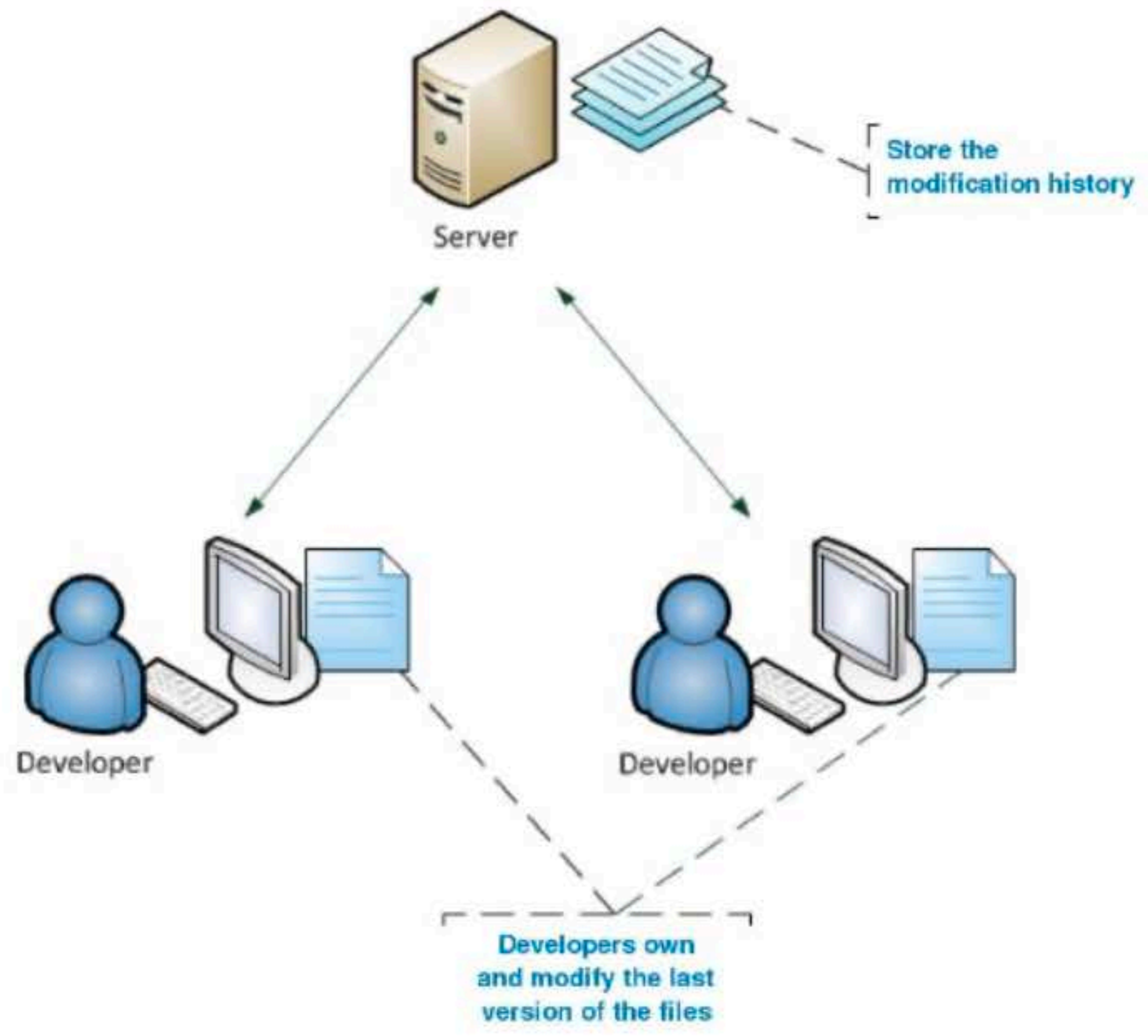
Version control

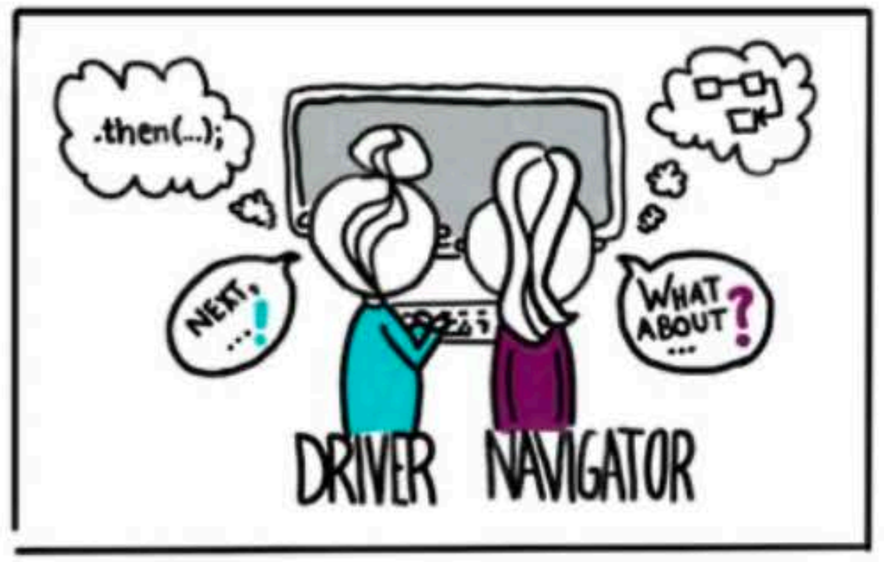


Continuous integration



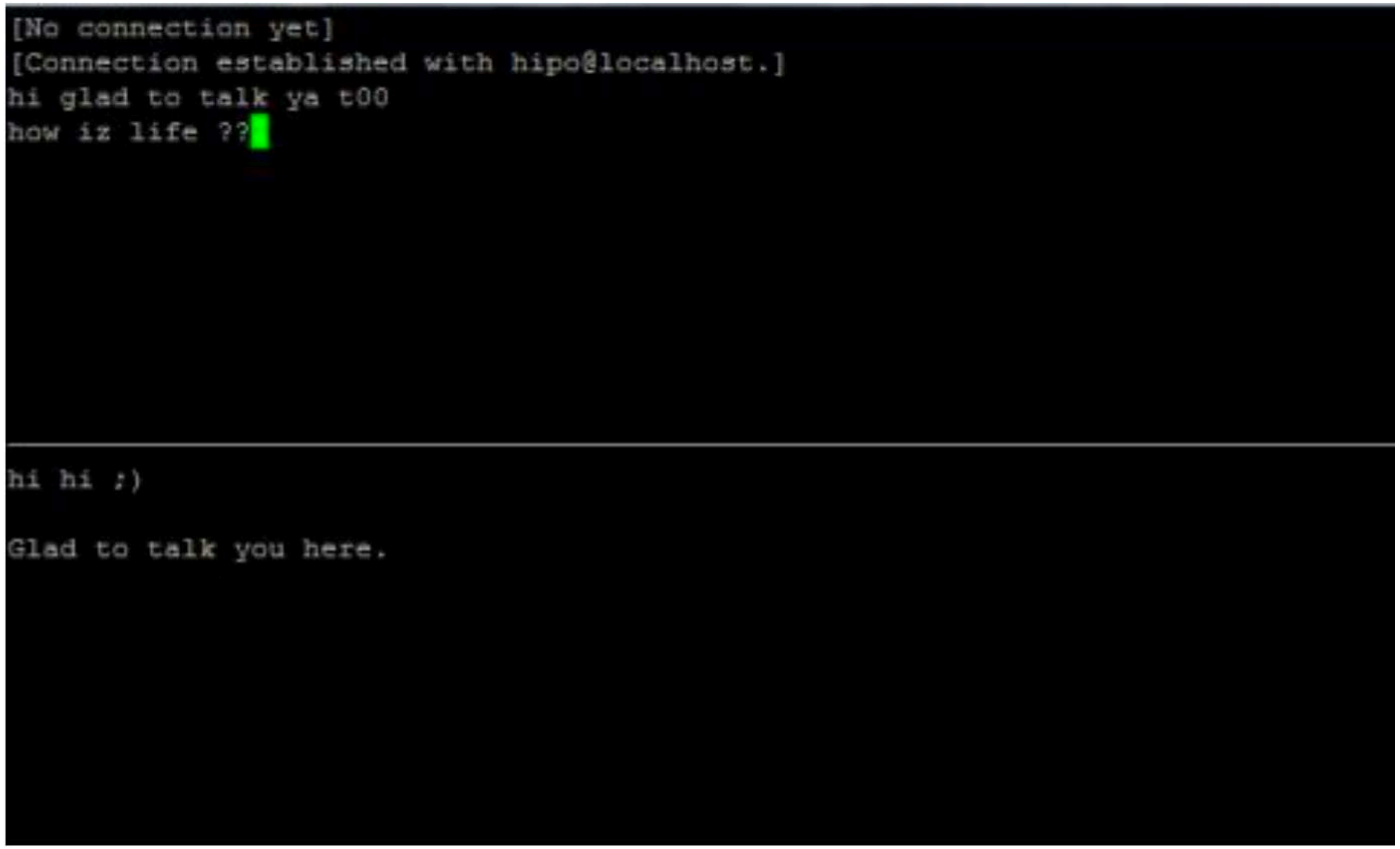
Agile development methods





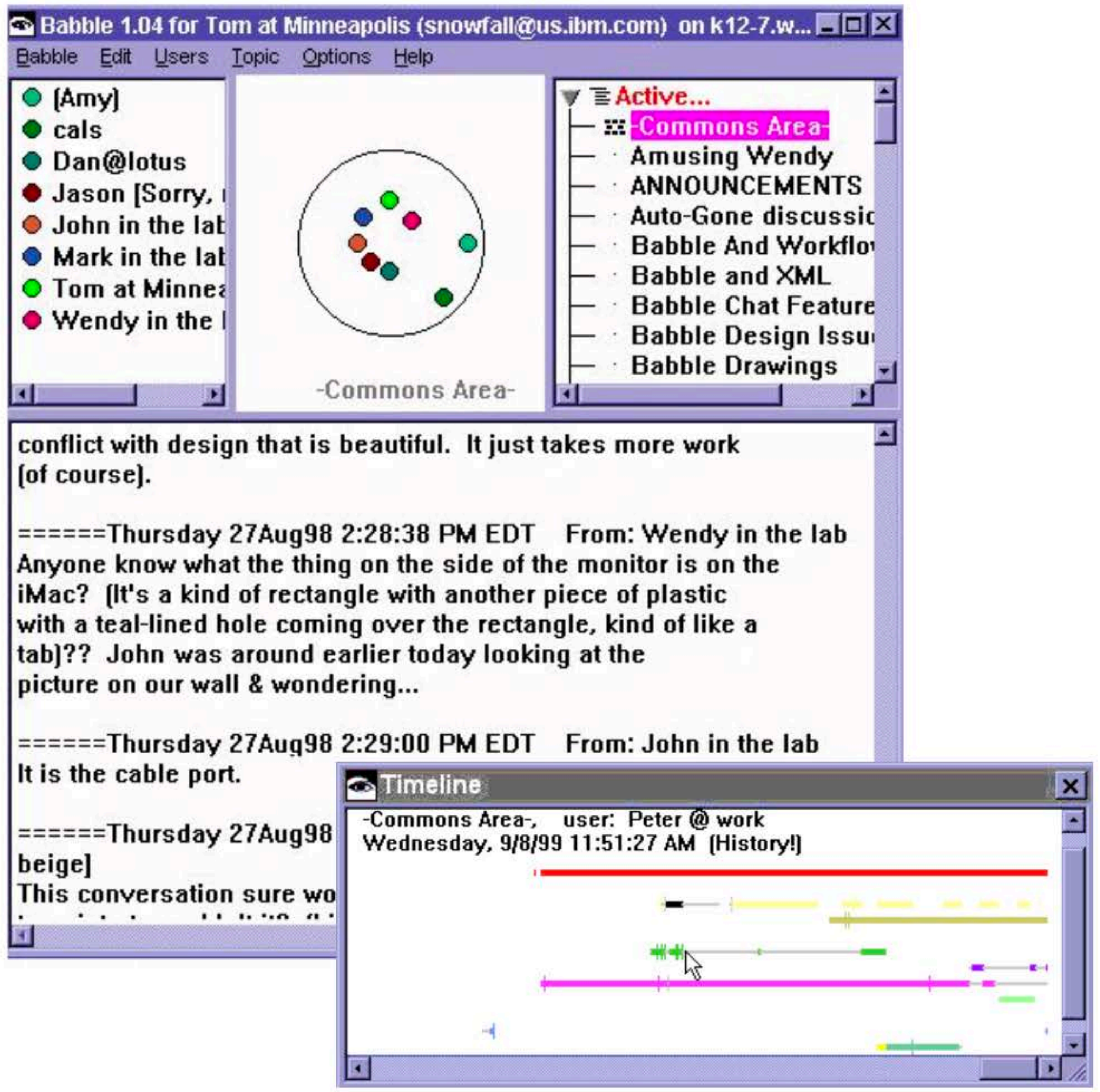
Communicate

Text-based real-time communication

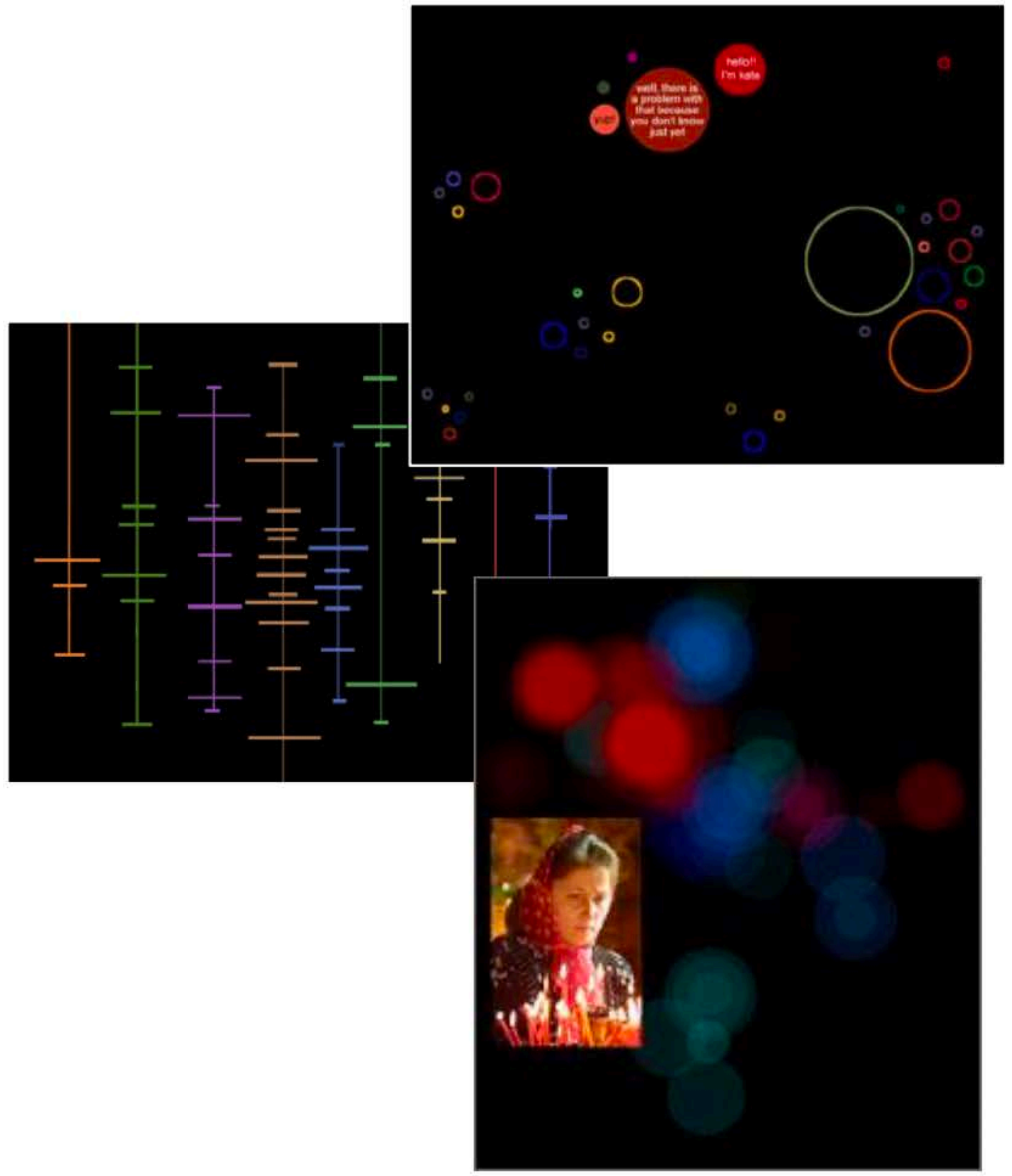


Unix talk

Chat rooms: for small groups



Babble (Bradner et al., 1988)
http://www.research.ibm.com/SocialComputing/babble.htm



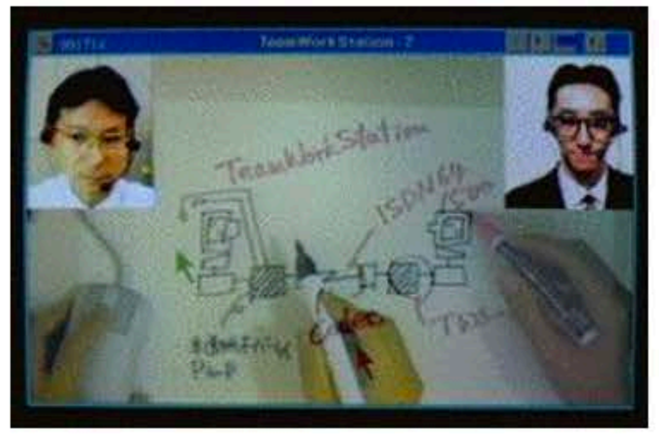
Chat circles (Viégas et al., 1999)
http://web.media.mit.edu/~fviegas/circles/
http://web.media.mit.edu/~fviegas/CC2/

Video-mediated communication

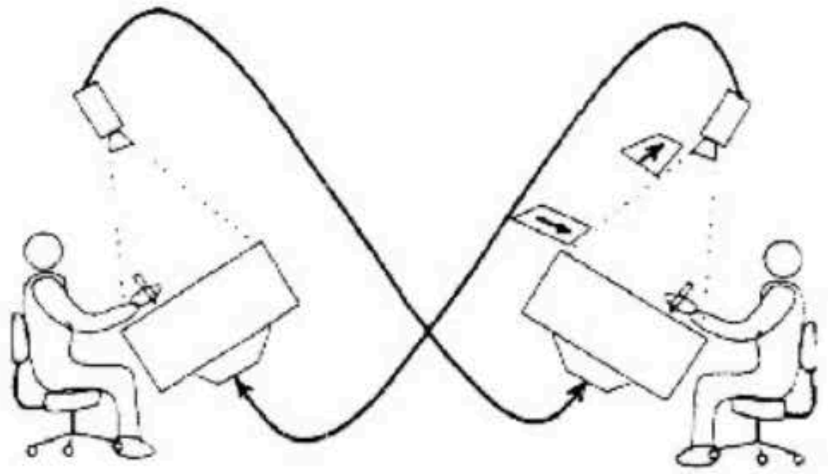
Hole-in-Space (1980)

Mediaspaces (1983-)

TeamWorkStation (1990)

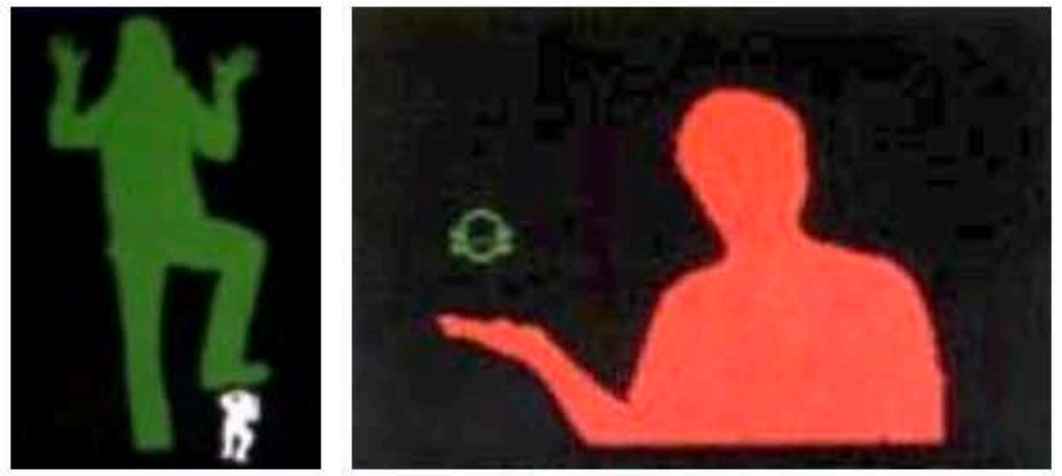


VideoDraw (1991)

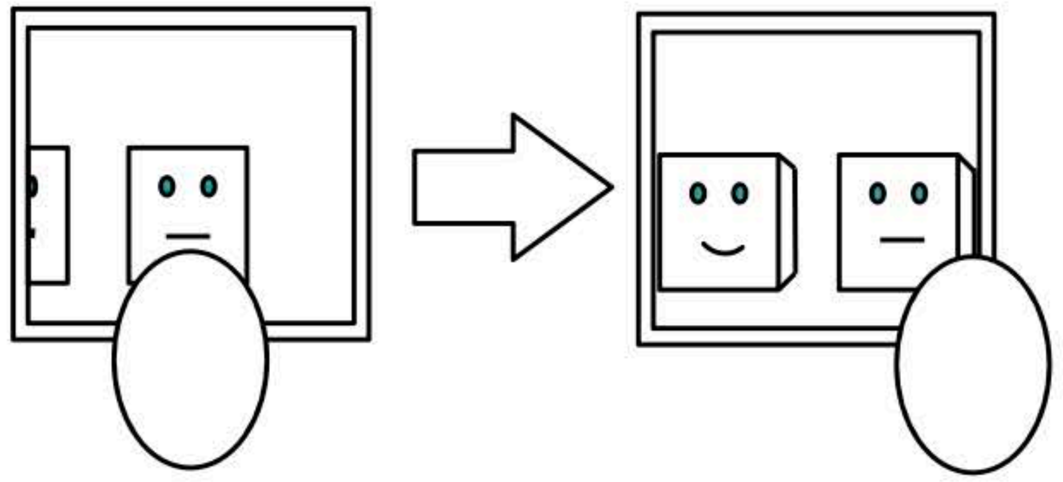


ClearBoard (1991-94)

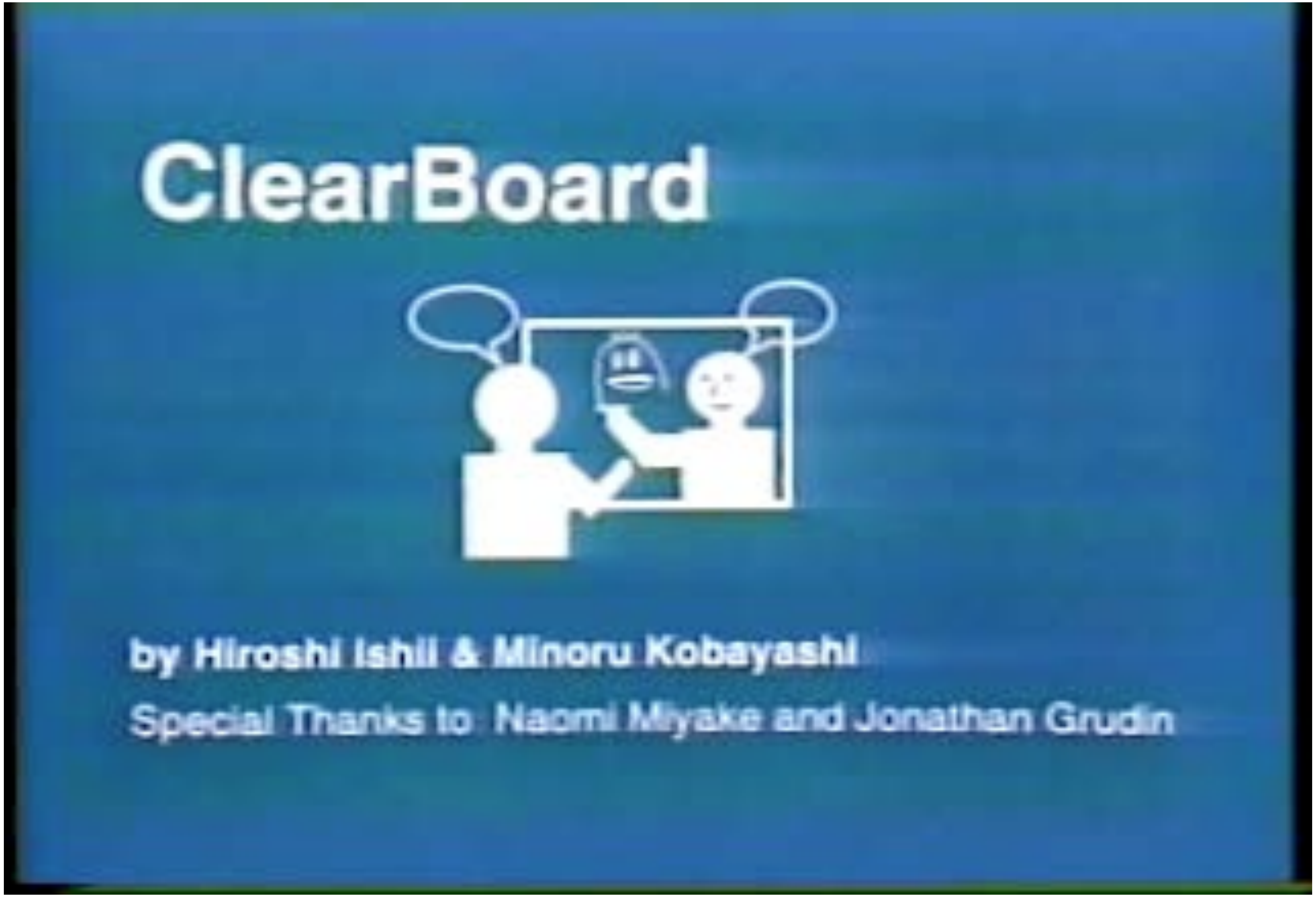
Videoplace (1974-85)



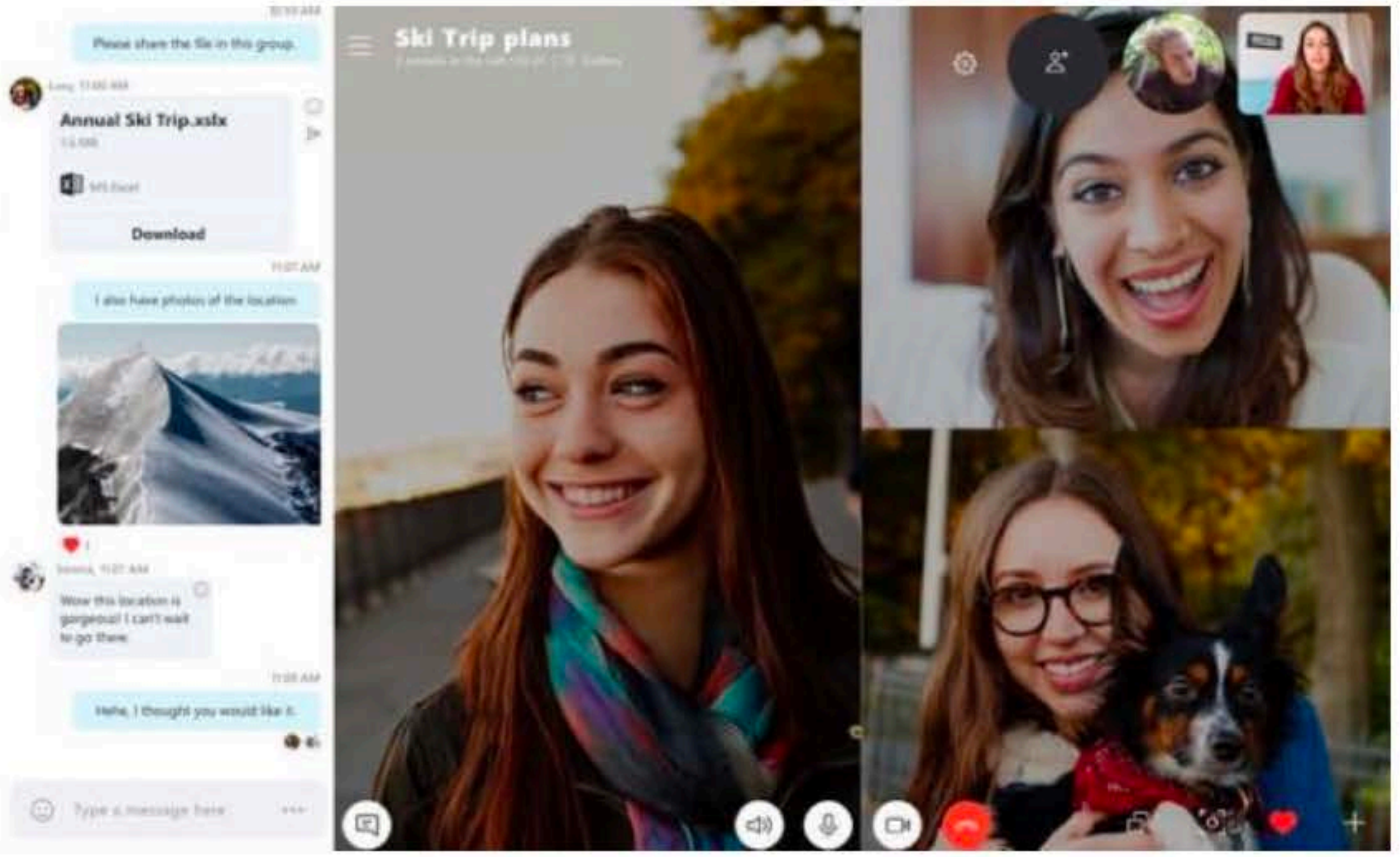
Virtual window (1995)



Clearboard


ClearBoard
by Hiroshi Ishii & Minoru Kobayashi
Special Thanks to Naomi Miyake and Jonathan Grudin

Videoconferencing: all the same



Skype

Zoom

Teams

Webex

Social networks: scale up

From small, “intimate” social networks to large groups of
- “followers”
  - Explosion of the number of users

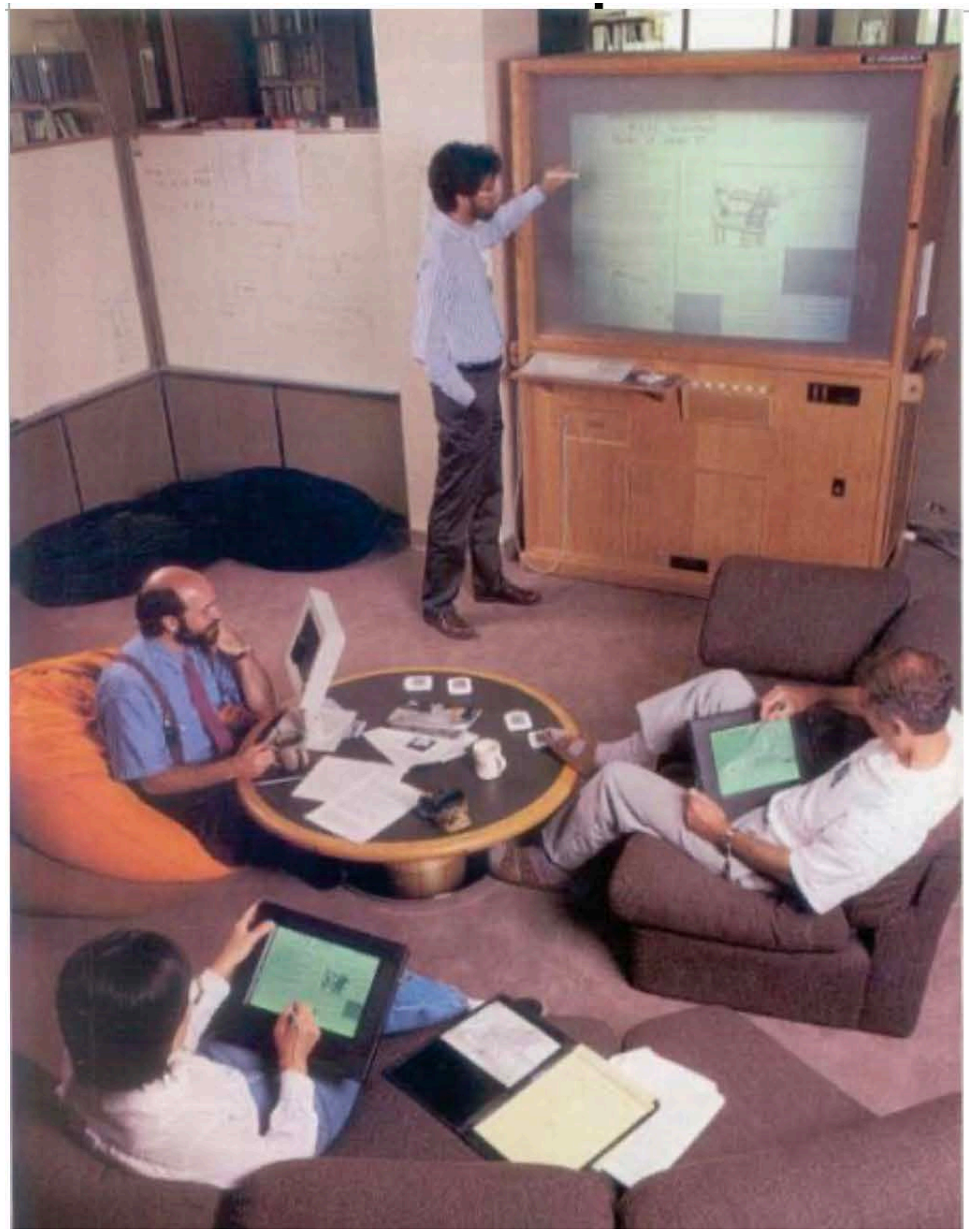
Shift towards pushing information to sustain “engagement”
- for better (Arab springs, …)
- and for worse (fake news, harassment, …)

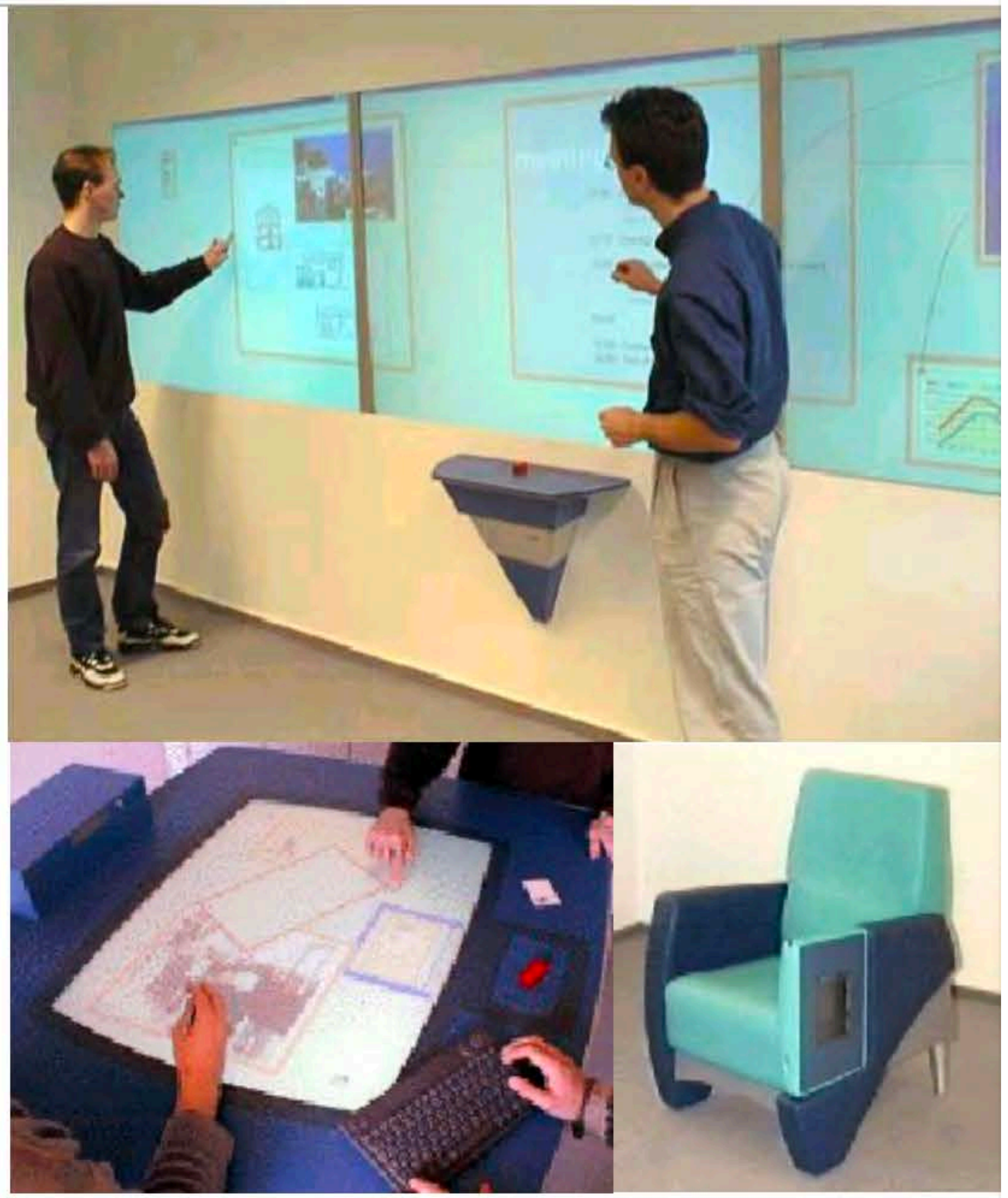
The fight for user attention to sustain the business model
- Large-scale experiments without informed consent
- Evidence that social media affects teens’ self-esteem

Regulate, but how?

Collaborate

Collaborative spaces





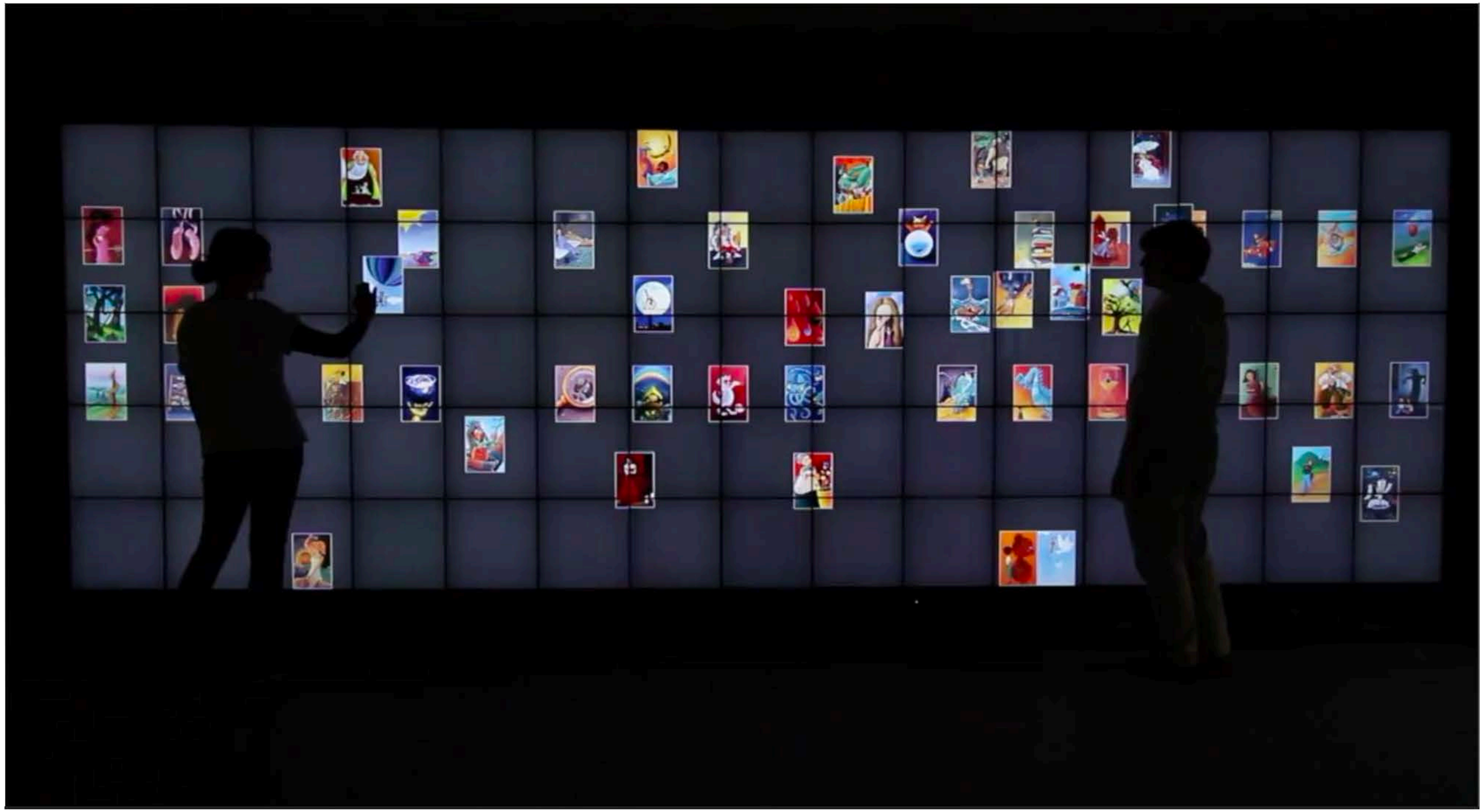
Cooperative buildings
(Streitz et al., 1998)

Ubicomp (Weiser, 1991)

Interactive spaces

Collaborate on a wall-sized display

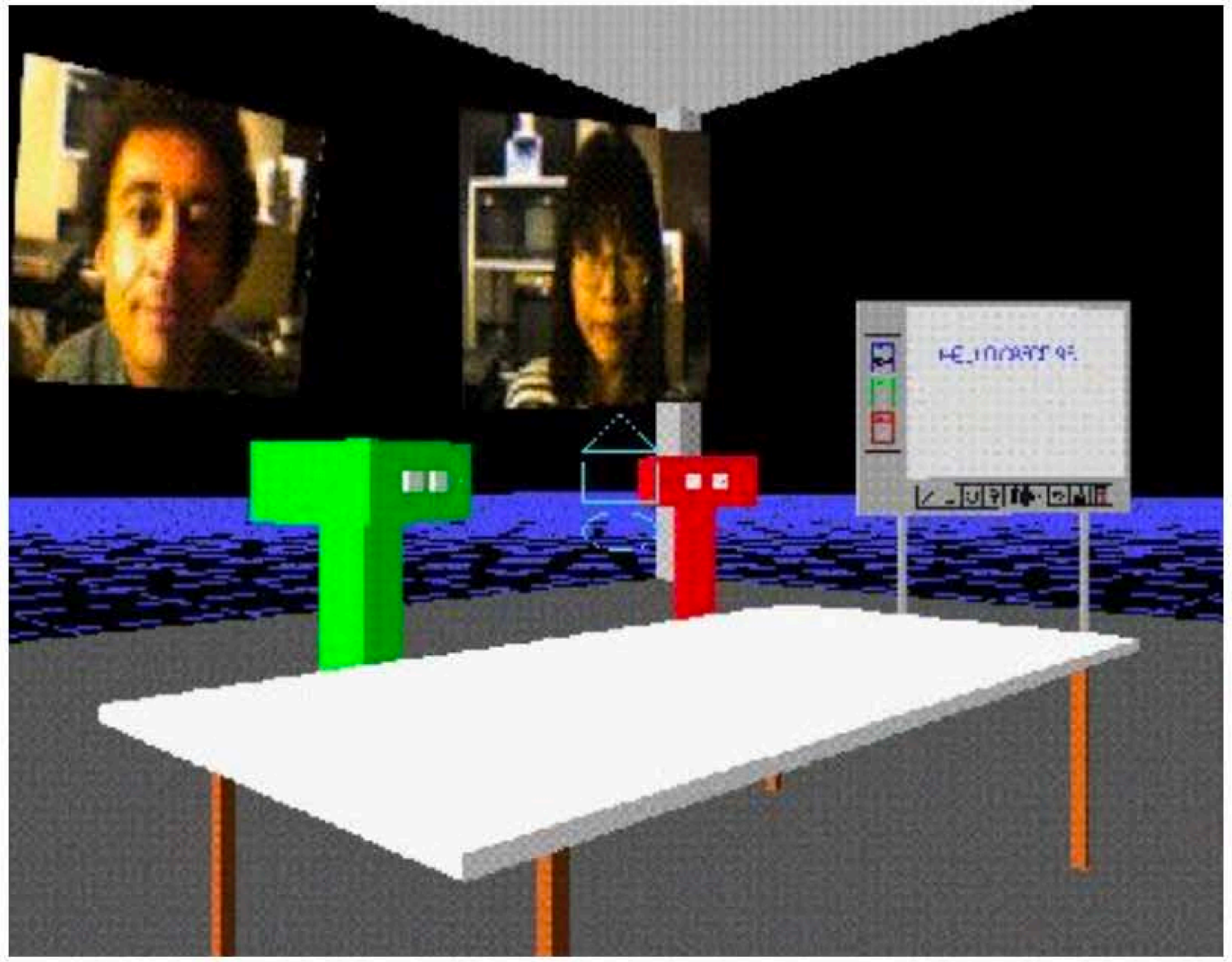
- CoReach: collaborative gestures (Liu et al., 2017)



https://www.youtube.com/watch?v=IvbEdic36PU

Collaborative Virtual Environments

Represent participants by avatars in a virtual world



DIVE (1991)

Second Life (2005)

Networked games

Real-time massively multiplayer games

World of Warcraft (2004)

Fortnite (2017)

The metaverse: utopia or dystopia?

Challenges for designers

- Who does the work vs. who gets the benefit
- Critical mass and Prisoner’s dilemma problems
- Disruption of social processes
- Exception handling
- Unobtrusive accessibility
- Difficulty of evaluation
- Failure of intuition
- Careful adoption process

Jonathan Grudin

Privacy, and other social behaviors

"On the Internet, nobody knows you're a dog."

Plausible deniability